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使用全腦分割影像提升神經膠質瘤復發樣態預測準確度
Enhancing the Prediction Accuracy of Glioblastoma Recurrence
Patterns Using Whole Brain Segmentation Images

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Chinese Abstract

本研究旨在開發一個基於放射組學特徵與深度學習技術的神經膠質瘤（GBM）復發樣態預測模型。研究資料來自基隆長庚紀念醫院 106 名患者的術前 MRI 影像。這些影像通過 Cancer Imaging Phenomics Toolkit (CaPTk) 的 BraTS Pipeline 進行預處理，包括 N4 偏差場校正和 Z 分數正規化，並標準化為 240x240x155 的大小。隨後，使用 FreeSurfer 的序列自適應多模態分割 (SAMSEG) 演算法將影像分割成多個大腦區域，用於分別產生放射組學特徵。考慮了各種類型的特徵，在特徵選擇過程中，採用了支援向量機-遞歸特徵消除 (SVM-RFE) 技術來識別最具預測力的特徵子集。選定的特徵用於訓練多層感知器 (MLP) 神經網路模型，並進行了 5 折交叉驗證以確保模型的泛化能力。具體而言，使用 256 個腦室特徵的模型達到了 98.14% 的預測準確率，而結合腦腫瘤和腦室特徵的模型則達到了 99.05% 的準確率。此外，還使用 Shapley Value 計算特徵重要性，提供了各個特徵對預測結果貢獻的見解。本研究不僅提高了預測 GBM 復發模式的準確性，還為臨床實踐提供了寶貴的決策支持，有助於制定更精確的治療策略。

關鍵字：神經膠質瘤、腦室、腦腫瘤復發樣態、醫學影像、支援向量機遞歸特徵消除、多層感知器

Abstract

This research aims to develop a model for predicting glioblastoma (GBM) recurrence patterns using radiomic features and deep learning techniques. Data were collected from preoperative MRI images of 106 patients at Keelung Chang Gung Memorial Hospital. These images were preprocessed using the Cancer Imaging Phenomics Toolkit (CaPTk) BraTS Pipeline, which included N4 bias field correction and Z-score normalization, and standardized to a size of 240x240x155. Subsequently, FreeSurfer's Sequence Adaptive Multimodal SEGmentation (SAMSEG) algorithm was used to segment the images into several brain areas, which is used to generate radiomic features separately. Various types of features were considered, and during the feature selection process, the Support Vector Machine-Recursive Feature Elimination (SVM-RFE) technique was used to identify the most predictive subset of features. The selected features were used to train a Multilayer Perceptron (MLP) neural network model, which underwent 5-fold cross-validation to ensure its generalizability. Specifically, the model achieved a prediction accuracy of 98.14% using 256 ventricular features, and an accuracy of 99.05% when combining brain tumor and ventricular features. Additionally, Shapley Values were used to calculate feature importance, providing insights into the contributions of individual features. This research not only enhances the accuracy of predicting GBM recurrence patterns but also offers valuable decision support for clinical practice, aiding in the development of more precise treatment strategies.

Keywords: Glioblastoma, Ventricle, Brain Tumor Recurrence Pattern, Medical Image, SVM-RFE, Multilayer perceptron



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Chapter 1 Introduction

Glioblastoma (GBM), as the most prevalent primary malignant brain tumor in adults, receives significant attention in both clinical treatment and research domains [1]. This tumor's aggressive nature and rapid growth make conventional treatment approaches largely ineffective, with a combined regimen of surgical resection, radiotherapy, and chemotherapy resulting in a median survival of only about 15 months for patients [2].

Moreover, the recurrence of GBM is almost inevitable, exhibiting various patterns ranging from local to extensive and even distant progression [3]. Studies indicate that most tumor recurrences occur within 2 centimeters of the resection margin, some recur within the radiotherapy target zone, and only a minority manifest as distant recurrences.

Different recurrence patterns significantly impact patient prognosis, with local recurrences generally having a better outlook than distant ones [4]. Thus, accurately predicting the recurrence patterns of GBM is crucial for guiding clinical treatment, extending patient survival, and enhancing quality of life.

Radiomics, an innovative technology in medical imaging analysis in recent years, offers new perspectives by extracting high-dimensional features from clinical imaging to reveal the molecular biological characteristics of tumors and predict disease progression [5][6].

With advanced technologies such as machine learning and deep learning, radiomics has demonstrated unprecedented potential in cancer diagnosis, treatment response evaluation, and prognosis prediction [7]. Particularly in the precise prediction of recurrence patterns of glioblastoma, efficiently integrating radiomics features with deep learning algorithms and handling large, high-dimensional datasets represent significant current challenges [6] [7].

Despite breakthroughs in medical imaging analysis with deep learning technologies and the emerging potential of radiomics to improve predictions of GBM recurrence, many studies have begun to explore predicting or identifying recurrence patterns of glioblastoma [5] [6] [7] [8]. However, current research still faces challenges in accurately segmenting tumor tissues from complex MRI images and effectively predicting their recurrence patterns. Additionally, traditional recurrence monitoring relies on MRI scans and subjective physician assessments, a time-consuming process that may delay recurrence detection and limit the possibility of timely and effective intervention.

This research aims to develop a new analytical method and predictive model by focusing on the significance of ventricle features in predicting the recurrence patterns of GBM. By integrating radiomic features with deep learning technologies and emphasizing brain structure, particularly the

ventricles, we aim to enhance the accuracy and efficiency of recurrence monitoring.

This approach seeks to provide effective decision support for clinical medicine, ultimately improving patient prognosis and quality of life.

This thesis is organized into six chapters:

- Chapter 1: Introduction - Outlines the clinical challenges of GBM, the significance of radiomics combined with deep learning, and the innovative aspects of the study.
- Chapter 2: Related Work - Reviews glioblastoma, the ventricular system, MRI in glioblastoma, brain tumor segmentation, CaPTk, FreeSurfer and SAMSEG, radiomics and feature extraction, SVM-RFE, and K-fold cross-validation.
- Chapter 3: Materials and Methods - Describes the research design, experimental environment, data collection and processing, feature extraction, feature selection, and model structure.
- Chapter 4: Experimental Results - Presents the performance of the predictive models for GBM recurrence patterns.
- Chapter 5: Discussion - Analyzes the clinical significance of the model's results and their impact on GBM treatment decisions.
- Chapter 6: Conclusion - Summarizes the main findings and suggests future research directions.

Chapter 2 Related Work

2.1 Glioblastoma and the Ventricular System

2.1.1 Glioblastoma

Glioblastoma Multiforme (GBM) occupies a profoundly significant position among adult primary brain tumors and is highly prioritized in clinical treatment and oncological research. This prominence is due not only to GBM being the most common and aggressive tumor within the central nervous system but also because of the substantial survival challenges it imposes on patients [1]. Clinically, GBM is characterized by rapid tumor growth and low responsiveness to current treatments, resulting in extremely limited therapeutic options and a median survival time of merely 15 months for patients [2]. Brain tumors result from the proliferation of abnormal cells within the brain. Based on their growth rate and the potential for recurrence post-treatment, brain tumors are categorized into malignant and benign types. Benign tumors are non-cancerous, grow slowly, and are less likely to recur after treatment.

Malignant tumors, however, are primarily composed of cancer cells capable of invading surrounding tissues or metastasizing to other body parts. Gliomas arise from mutations in glial cells, which transform normal cells into malignant ones, and they represent the most common form of

astrocytoma, accounting for 30% of all brain and central nervous system tumors and 80% of all malignant tumors [17]. Figure 2.1 shows an example MRI of LGG and HGG.

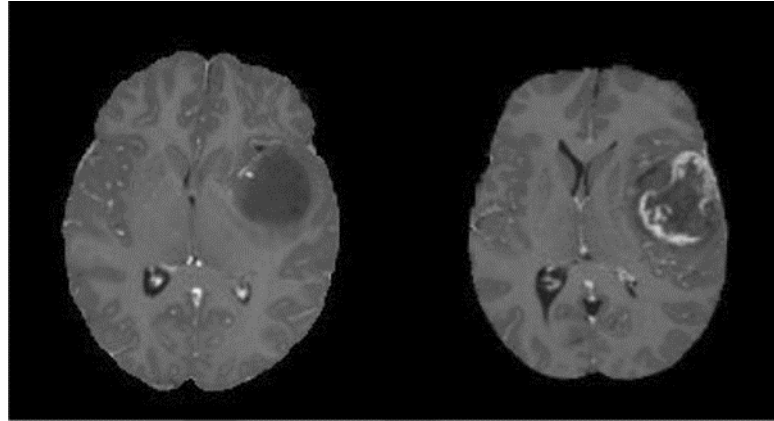


Figure 2.1 An example of an MRI showing low-grade glioma (LGG, on the left) and high-grade glioma (HGG, on the right). Data sourced from BraTS 2019 [17].

The phenotypical makeup of gliomas can include astrocytomas, oligodendrogliomas, or ependymomas, each displaying distinct behaviors. The World Health Organization (WHO) utilizes a grading-based methodology to classify each tumor according to its level of aggressiveness [17]:

- Grade I tumors are typically benign, mostly curable, and commonly observed in children.
- Grade II tumors encompass astrocytomas, oligodendrogliomas, and mixed oligoastrocytomas, which are more prevalent in adults. All low-grade gliomas may eventually progress to high-grade tumors.

- Grade III tumors may consist of anaplastic astrocytomas, oligodendrogliomas, or mixed oligoastrocytomas, exhibiting greater aggression and infiltration than grade II tumors.
- Grade IV tumors, also known as Glioblastoma Multiforme (GBM), are the most aggressive form within the WHO classification.

Grades I and II gliomas are considered low-grade, while grades III and IV are classified as high-grade gliomas [1]. Low-grade gliomas (LGG) are benign and can be surgically resected. In contrast, high-grade gliomas (HGG) are malignant, challenging to excise surgically due to their extensive invasion into surrounding tissues.

GBM is the most common and aggressive form, classified as a grade IV tumor, underscoring its characteristics as an extremely malignant neoplasm [1]. The challenge in treating GBM is partly attributed to its significant biological heterogeneity. The World Health Organization's CNS5 has recently reclassified gliomas into six distinct families, including adult-type diffuse gliomas, pediatric-type diffuse low-grade gliomas, pediatric-type diffuse high-grade gliomas, circumscribed astrocytic gliomas, glioneuronal and neuronal tumors, and ependymomas [1].

Moreover, the intertumoral and intratumoral heterogeneity of GBM manifests in genetic variations, such as IDH1/2 mutations, 1p/19q co-deletion, and the methylation status of the MGMT gene, all of which are

key molecular markers closely associated with patient prognosis and response to chemotherapeutic agents [3].

Another significant hurdle for GBM patients is the presence of the blood-brain barrier (BBB), which notably restricts the ability of most potentially effective therapeutic drugs to penetrate the brain [11]. This increases the difficulty of treating GBM and makes it particularly important to develop new treatment strategies that can cross the BBB [9][11].

The clinical progression patterns of GBM are varied, including local, distant, diffuse, or multifocal progression, all of which critically influence treatment decisions and prognostic evaluations [3].

Additionally, the heterogeneity among and within tumors is reflected not only in their genetic and phenotypic characteristics but also in their proteomic features, which are crucial for understanding the complexity of GBM and developing new therapeutic targets [10].

Given these attributes of GBM, integrating the latest imaging technologies with deep learning algorithms is of paramount importance for enhancing the diagnostic accuracy of GBM, predicting treatment response, and monitoring disease progression. This interdisciplinary approach opens up new avenues for improving treatment outcomes and prognoses for patients with GBM.

2.1.2 Ventricular System

The ventricular system is a critically important and complex neural structure in the human brain. Situated within the brain matter, it consists of spaces with specific functions and locations.

Specifically, the cerebral hemispheres house the left and right lateral ventricles on either side of the brain. Deeper within the brain, the third ventricle is located in the thalamus, while the fourth ventricle lies between the cerebellum and the medulla oblongata, bridging the gap to the pons.

Connecting the third and fourth ventricles is a slender channel known as the cerebral aqueduct, which facilitates the flow of cerebrospinal fluid (CSF) between these two ventricles [12]. Figure 2.2 is a schematic diagram of the structure of the brain ventricles.

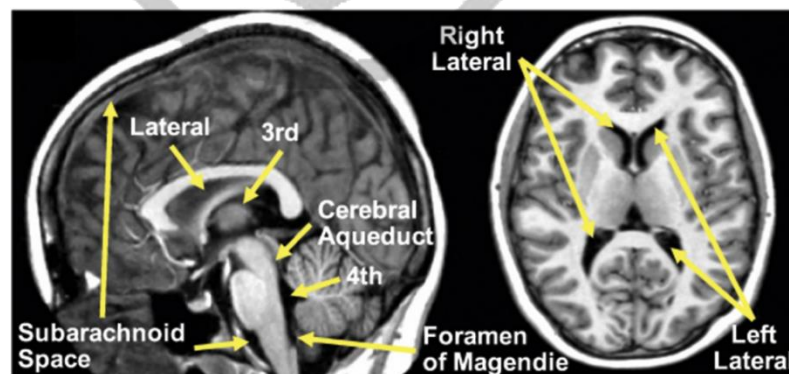


Figure 2.2 An example of an MRI image of brain ventricle architecture [12].

The ventricles contain the choroid plexus, a vascular-rich structure primarily responsible for CSF production [15]. CSF is a clear, protective

fluid that, once secreted by the choroid plexus, fills the entire ventricular system and eventually flows into the subarachnoid space [14]. In the subarachnoid space, the CSF surrounds the brain and spinal cord, providing nutrition and nourishment and acting as a buffer to protect these sensitive neural tissues from external shocks and internal pressure fluctuations [12].

The production, flow, and absorption of CSF are finely regulated processes, and any abnormalities in this system, such as blockages in the ventricles, can lead to obstructed CSF flow. When an obstruction occurs in any part of the ventricular system, CSF can accumulate within the ventricles, leading to a condition known as obstructive hydrocephalus [14]. This condition increases intracranial pressure, compresses brain tissue, and can also negatively affect the patient's cognitive functions, physical coordination, and overall health [12].

Historically, the ventricles have occupied a significant place in medicine and philosophy, attracting the attention of philosophers, clergy, anatomists, and physicians. Since the time of Aristotle, the ventricles were initially thought to house the soul and were responsible for higher functions as the "vessels of life." Influenced by Christianity and the Renaissance, the concept of ventricles evolved into chambers assigned with common sense, creative imagination, and memory. It was not until the 16th century that anatomists such as Andreas Vesalius and Costanzo Varolio clearly stated

that the ventricles were filled with cerebrospinal fluid. The previous view was gradually replaced by the scientific method, and the function of the ventricles was understood more as related to the cerebrospinal fluid [13].

Before the advent of modern medical imaging techniques, techniques like ventriculography and pneumoencephalography were used to study the morphology of the ventricles to assess brain health. Today, asymmetry and anatomical variations of the ventricles are commonly observed in imaging and are generally considered normal variations. Although these variations usually do not pose diagnostic difficulties, medical imaging interpreters remain vigilant to avoid mistaking normal variations for pathological changes [13]. Pathological conditions such as hydrocephalus, degenerative diseases, or traumatic brain injury can affect the size and shape of the ventricles [14]. With modern imaging diagnostics like MRI and CT scans, physicians can closely observe the morphology of the ventricles, identify potential abnormalities, and devise appropriate treatment plans [13].

2.2 MRI Analysis and Brain Tumor Segmentation

2.2.1 MRI in Glioblastoma

In the field of medical imaging analysis for glioblastoma, MRI technology plays a pivotal role. Image segmentation aims to partition the imaging into meaningful segments, thereby identifying critical structures

or areas. For glioblastoma, accurate segmentation is essential for precise diagnosis and the formulation of effective treatment strategies. Various MRI sequences, including T1-weighted (T1), and contrast-enhanced images (T1C), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR), provide unique insights into the tumor and its surrounding tissues [16]. These imaging details are crucial for accurately pinpointing the tumor core and devising surgical strategies.

In brain MRI scans, T1 images provide distinct contrast between gray and white matter, whereas T1 and T1c images are crucial for evaluating changes in tumor shape and the enhanced delineation of tumor margins. T2 images prominently reveal fluid presence separate from cortical tissue, and lesions are clearly outlined on FLAIR images [18].

The T1 scan distinguishes between gray and white matter, with T2 being sensitive to water content, making it suitable for conditions involving water accumulation in brain tissue [18]. Both T1 and T2 images are employed to differentiate cerebrospinal fluid (CSF)—a colorless fluid found in the brain and spinal cord, appearing dark in T1 imaging and bright in T2 imaging. FLAIR, the fourth sequence, is akin to a T2 image but with a different acquisition protocol. It is utilized in pathology to differentiate between CSF and brain irregularities, capable of localizing edema regions from CSF by suppressing free water signals, thereby rendering

periventricular hyperintense lesions vividly visible in the images [19].

Through these advanced MRI techniques, physicians can closely observe the tumor's growth pattern, the response of surrounding tissues, and changes before and after treatment, which are invaluable for guiding clinical management. Consequently, MRI holds an indispensable role in the diagnosis and treatment planning of glioblastoma, with ongoing technological innovations and research further enhancing its clinical application value. As shown in Fig 3, it is four different MRI images.

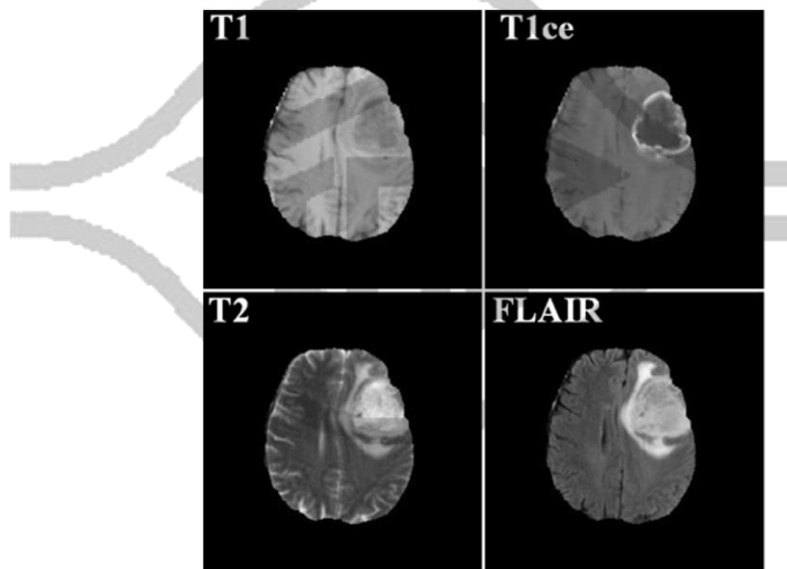


Figure 2.3 Depicts four different types of MRI images: T1, T2, T1ce, and FLAIR [16].

2.2.2 Brain Tumor Segmentation

According to [16], the diagnosis and therapeutic assessment of gliomas, MRI imaging provides rich anatomical and pathological information. Correct identification and classification of different tumor components are

critical for disease assessment and the formulation of treatment plans. Figure 2.4 showcases the three key regions within a tumor on MRI images.

The Enhancing Tumor (ET), the Tumor Core (TC), and the Whole Tumor (WT), along with their clinical significance [16]:

- Enhancing Tumor (ET): Panel A is the enhancing region is highlighted in yellow, representing the area of the tumor with the most contrast uptake, typically associated with the most active and invasive parts of the tumor.
- Tumor Core (TC): Panel B presents the tumor core outlined in magenta, encompassing the enhancing tumor regions as well as necrotic tissue, omitting the edematous area. The accurate segmentation of the tumor core is crucial for pre-surgical planning and assessment of therapeutic responses.
- Whole Tumor (WT): Panel C illustrates the whole tumor in cyan, which includes the tumor core and surrounding edema/invasion areas, reflecting the full extent of the tumor's spread.

Panel D displays the combination of segmented tumor subregions, providing an insightful representation of the overall tumor structure essential for the training and evaluating algorithms. The regions - edema/invasion (green), enhancing tumor (blue), and necrotic (red) - are distinctly segregated and labeled in the three-dimensional space[16].

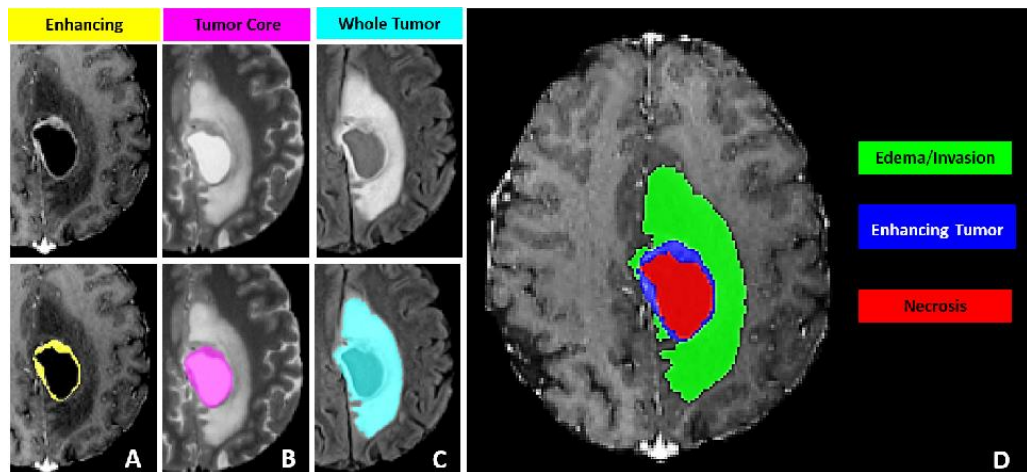


Figure 2.4 BraTS 2021 challenge, schematic diagram of brain tumor segmentation [16].

The identification of these segmented areas within automated imaging segmentation tools is critical and pivotal for radiomic feature extraction from the images. These features will be used to build predictive models further applied for assessing the treatment effects on brain tumors and predicting the risk of recurrence, which is an important aspect of precision medicine. This section will further explore the identification of these subregions from multiparametric MRI scans and how this structural information can be leveraged to enhance the accuracy of tumor diagnostics, treatment monitoring, and prognostic evaluations [16].

2.3 Medical Imaging Segmentation Tools

2.3.1 Cancer Imaging Phenomics Toolkit

The Cancer Imaging Phenomics Toolkit (CaPTk) is a beacon in the field of cancer imaging analytics, bringing forth a multifaceted software platform that bridges the gap between radiology and personalized medicine. As detailed in the technical overview by the Center for Biomedical Image Computing and Analytics at the University of Pennsylvania, CaPTk is a versatile, open-source toolkit that provides end-to-end solutions for cancer imaging phenomics research [21].

Through its comprehensive suite of tools, CaPTk enables researchers to perform an array of operations critical to cancer imaging analysis, such as segmentation, registration, and feature extraction. These processes are foundational to understanding the intricate patterns within cancer imaging data and are pivotal in developing robust predictive models for patient diagnosis and prognosis [21]. An illustration of CaPTk's two-level architecture and the extensive range of functions it offers is depicted in Figure 2.5, providing a visual representation of the toolkit's capabilities from image input to predictive modeling outcomes [22].

The toolkit is equipped with a user-friendly interface, ensuring that sophisticated analyses that once required extensive computational expertise are now accessible to a broader range of clinical researchers. This

democratization of advanced imaging analytics is critical, as it allows for a more inclusive progression in precision diagnostics and the predictive modeling of clinical outcomes [22].

One of CaPTk's key components, the BraTS Pipeline, exemplifies the toolkit's application in analyzing brain tumors, particularly glioblastoma, which is considered to be one of the most aggressive forms of cancer. The pipeline has been instrumental in advancing our understanding of glioblastoma's behavior and response to therapies [21].

Furthermore, CaPTk's role in quantitative imaging analytics for precision diagnostics, as explored in various studies [22], underscores its significance in not only brain tumors but in various types of cancer, manifesting its adaptability and effectiveness.

The interface of CaPTk, which allows for the seamless execution of the aforementioned processes, is showcased in Figure 2.6. This interface provides researchers with a straightforward pathway to implement complex analyses, underscoring the toolkit's emphasis on usability and efficiency.

Despite the transformative potential of CaPTk, it is important to recognize the challenges inherent in its adoption. The intricate nature of the toolkit's functions may present a steep learning curve for new users. Moreover, the computational demands of some features necessitate substantial processing power, which can be prohibitive for institutions with

limited resources. Nevertheless, CaPTk continues to evolve, with ongoing efforts to enhance its computational efficiency and expand its utility.

As CaPTk continues to integrate more advanced artificial intelligence and machine learning algorithms, its future prospects in the realm of radiogenomics and cancer treatment planning are incredibly promising. The toolkit is poised to play a crucial role in the advent of novel radiomic signatures and molecular characteristics that could revolutionize cancer diagnostics and prognostics [22].

By facilitating a standardized approach to feature extraction and analysis, CaPTk contributes significantly to the convergence of clinical radiology with precision medicine, potentially leading to improved clinical outcomes.

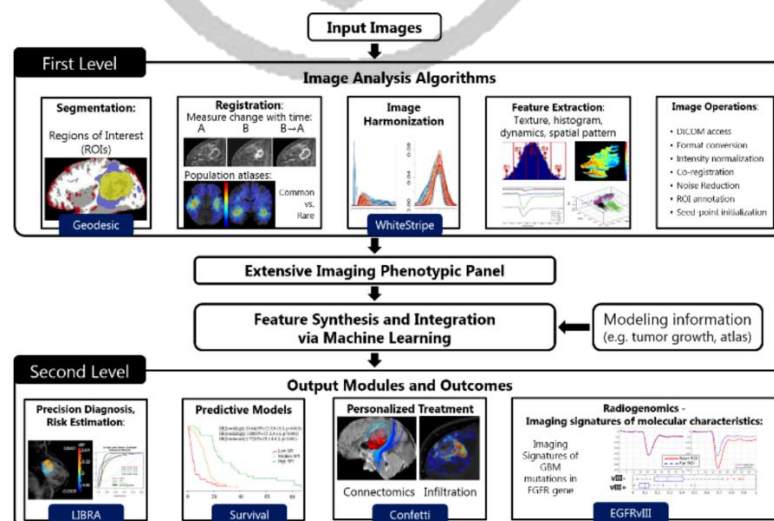


Fig.1. Overview of all functions and applications of CaPTk, in its two-level architecture

Figure 2.5 Overview of all functions and applications of CaPTk [22]

In conclusion, CaPTk stands as a paradigm of innovation in cancer imaging, encapsulating the essence of precision medicine. Its applications extend across diverse research scenarios, making it an invaluable asset in the quest to unravel the complexities of cancer and offering a ray of hope in the personalized treatment of this challenging disease.

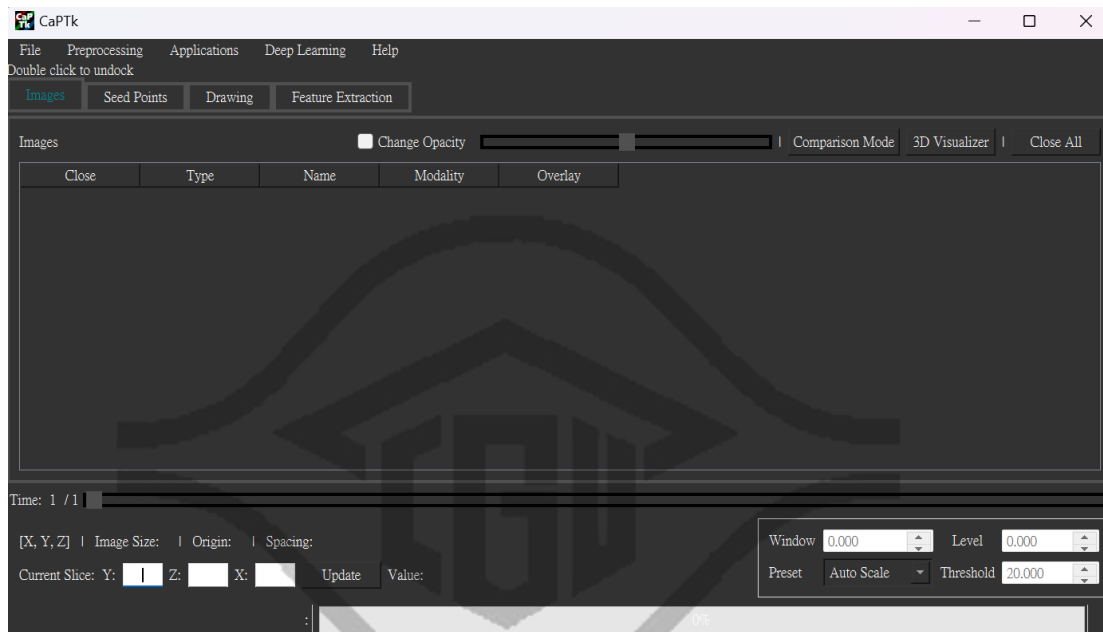


Figure 2.6 GUI of CaPTk

2.3.2 FreeSurfer and SAMSEG

Segmentation of brain structures using magnetic resonance imaging (MRI) is a cornerstone in neuroscientific research and critical in clinical diagnostics. The ability to precisely delineate various brain compartments facilitates a deeper understanding of neurological disorders, aids in surgical planning, and is essential for assessing responses to treatments. In this

regard, FreeSurfer has established itself as an indispensable tool in the field of neuroimaging. It offers a robust set of algorithms for cortical reconstruction and volumetric segmentation, providing detailed insights into the brain's anatomy and functionality [23].

FreeSurfer excels in the detailed mapping of cortical thickness, surface area, and volume, enabling researchers to explore brain morphology across different patient populations and disease states. Its capabilities extend to longitudinal analysis, where changes in brain structure over time can be meticulously tracked, offering invaluable data for studying progressive neurological conditions [23].

Complementing the capabilities of FreeSurfer, the SAMSEG (Sequence Adaptive Multimodal SEGmentation) algorithm represents a significant advancement in segmentation technology. Developed to enhance the adaptability and efficiency of brain tissue segmentation, SAMSEG utilizes a sophisticated parametric Bayesian model to adapt to the diverse sequence characteristics inherent in MRI data [20]. This model adjusts dynamically to the imaging parameters, allowing for consistent and precise segmentation across various imaging conditions without manual recalibration.

SAMSEG's integration with FreeSurfer leverages both tools' strengths, providing a seamless workflow for the segmentation of brain structures.

The synergy between FreeSurfer's detailed anatomical segmentation and SAMSEG's adaptability to MRI sequence variations facilitates a more accurate and comprehensive analysis of brain imaging data. This integration is particularly beneficial in multi-center studies, where data variability can pose significant analytical challenges [20].

The combination of FreeSurfer and SAMSEG addresses several key challenges in brain imaging. For instance, in clinical settings such as the treatment monitoring of glioblastoma, precise segmentation of tumor margins is crucial. The adaptive nature of SAMSEG, when combined with the anatomical detail provided by FreeSurfer, enables clinicians to delineate tumor boundaries accurately, enhancing the effectiveness of surgical and therapeutic interventions [20].

Furthermore, the robust statistical foundation of SAMSEG ensures that the segmentation is accurate and reproducible across different datasets and MRI protocols. This feature is essential for clinical trials and studies involving longitudinal data collection, where consistency in data interpretation is paramount.

In research contexts, the detailed brain maps generated by FreeSurfer, combined with the adaptive segmentation capabilities of SAMSEG, allow for sophisticated analyses of brain structure and function. Researchers can investigate complex patterns of brain development, aging, and disease

progression with greater precision. The insights gained from these studies contribute significantly to our understanding of neurodegenerative diseases, mental health disorders, and the impact of therapeutic interventions on brain structure and function [20] [23].

The ongoing development of FreeSurfer and SAMSEG highlights the dynamic nature of neuroimaging research. With continuous improvements, these tools are expected to offer even greater accuracy, faster processing times, and enhanced user-friendliness, making advanced neuroimaging techniques more accessible to researchers and clinicians worldwide. The integration of such tools into clinical practice is revolutionizing the diagnosis and treatment of neurological conditions, underscoring the importance of advanced image analysis in modern medicine [20][23].

By providing tools that combine detailed anatomical segmentation with adaptive, sequence-aware processing, FreeSurfer and SAMSEG empower the medical community to achieve more accurate diagnoses, tailor treatments to individual patients, and ultimately improve clinical outcomes. The integration of these technologies into neuroimaging workflows is a testament to the ongoing evolution of medical technology and its capacity to enhance our understanding of the human brain in health and disease [20][23].

2.4 Radiomics and Machine Learning Techniques

2.4.1 Radiomics and Feature Extraction

Radiomics is a transformative field within medical imaging, focusing on extracting and analyzing quantitative features from medical images. This approach converts traditional imaging data into a high-dimensional, quantifiable space that allows for the identification of patterns directly correlating with pathological features and clinical outcomes. As noted in [24], radiomics facilitates a nuanced depiction of the tumor environment, providing a rich dataset that supports precision medicine applications, particularly in managing complex diseases like glioblastoma. Moreover, the application of radiomics extends across various imaging modalities such as CT, MRI, and PET scans, highlighting its versatility in clinical diagnostics. This adaptability enables radiomics to provide comprehensive insights into disease characteristics across different imaging platforms, enhancing its utility in personalized medicine strategies.

The true potential of radiomics extends beyond image analysis, interfacing with clinical parameters to provide a more holistic view of the patient's condition. By correlating radiomic features with clinical outcomes, such as survival rates and therapeutic responses, researchers can uncover patterns that might not be visible through imaging alone. This integration enhances the predictive models, making them more robust and clinically

relevant.

In the realm of radiomics, the process of feature extraction is divided into several categories including first-order statistics, shape-based features, textural features, and higher-order statistical features. First-order statistics describe the distribution of pixel intensities within an image and include metrics like mean, median, entropy, and standard deviation. These features provide fundamental insights into the tumor's radiographic phenotype, with studies noted in [24] demonstrating their significant predictive value for disease outcomes.

Shape-based features assess the geometric properties of the tumor, crucial for understanding its growth patterns and potential invasiveness. Attributes such as size, shape, and surface irregularities are vital for tracking disease progression and evaluating therapeutic efficacy. Research discussed in [25] highlights the correlation of these features with tumor aggressiveness and prognosis.

The shape features include metrics such as volume, surface area, sphericity, and compactness, which offer insights into the morphological characteristics of the tumor and its evolution over time [30].

The extraction of textural features delves into the tumor's internal architecture, utilizing techniques such as the Gray Level Co-occurrence Matrix (GLCM), Gray Level Run Length Matrix (GLRLM), Gray Level

Size Zone Matrix (GLSZM), and Gray Level Difference Matrix (GLDM) to quantify tissue heterogeneity. These metrics are pivotal in uncovering variations in tumor pathology, providing insights into the tumor's aggressiveness and response to treatment [25].

Texture features capture spatial relationships between pixel intensities, revealing patterns that correspond to tissue heterogeneity and microenvironmental complexity [30].

Advancements in machine learning have propelled the capabilities of radiomic feature extraction, allowing for the development of sophisticated algorithms that can learn from data and identify patterns that human analysis cannot. These algorithms optimize the selection and combination of radiomic features, enhancing the predictive power of models used for diagnosing and prognosticating glioblastoma [30].

Higher-order statistical features utilize complex mathematical models like wavelet transforms to capture information from various image scales and orientations, thus providing a more detailed understanding of the tumor's architecture. This comprehensive approach is crucial for tailoring personalized treatment plans and enhancing the predictive accuracy of radiomic assessments. Wavelet features decompose images into different frequency components, enabling multi-scale analysis that captures both fine details and broad patterns [30].

The integration of these radiomic features into advanced machine learning models, as discussed in the critical reflection by [26], emphasizes the potential of radiomics to revolutionize clinical decision-making. These models not only classify and predict glioblastoma outcomes but also enhance our understanding of the biological processes driving tumor behavior. Machine learning techniques, including support vector machines, random forests, and neural networks, are employed to develop predictive models that leverage the rich radiomic feature sets [30].

Overall, the section on radiomics and feature extraction establishes the crucial role of this technology in advancing glioblastoma research and treatment. By systematically quantifying the complex characteristics of tumor imaging and employing advanced analytical models, radiomics proves to be an invaluable tool in refining diagnostics and optimizing therapeutic strategies. This comprehensive analysis, supported by the insights from [24] and the practical guide provided in [26], propels the frontiers of precision medicine for glioblastoma.

2.4.2 Support Vector Machine-Recursive Feature Elimination

In this section, we delve into the SVM-RFE (Support Vector Machine-Recursive Feature Elimination) technique, which is utilized extensively for feature selection in machine learning models applied to medical imaging,

as demonstrated in studies such as those by Guyon et al. [27] and Sanz et al. [28]. SVM-RFE employs a support vector machine framework to iteratively eliminate the least significant features iteratively, thereby focusing the model on the most influential features.

This method is particularly effective in enhancing the accuracy and interpretability of models by retaining only those features that significantly impact the outcome.

The application of SVM-RFE in the field of radiomics is crucial for optimizing the selection of diagnostic and prognostic features in complex diseases such as glioblastoma. By systematically pruning less critical features, SVM-RFE streamlines the analytical model and emphasizes features that are predictive of tumor behavior and patient outcomes. This selective process is key to developing targeted therapeutic strategies and advancing precision medicine, as evidenced by the refined feature selection methodologies discussed in [27] and the enhanced visualization techniques for feature relevance in [28].

Overall, SVM-RFE is instrumental in advancing machine learning applications within medical imaging. It provides a rigorous framework for feature reduction and selection, essential for developing highly accurate predictive models. This approach simplifies the modeling process and ensures that the features used are the most effective at predicting disease

characteristics and patient responses, thereby significantly influencing treatment decisions and improving clinical outcomes.

2.4.3 K-fold Cross-validation

K-fold cross-validation is a common technique in machine learning for assessing model performance and stability. This method involves partitioning the dataset into k equal-sized subsets, known as 'folds'. Each fold is used as a validation set once, while the remaining $k - 1$ folds are combined to form the training set. This process is repeated k times, ensuring that each fold serves as the validation set once. This iterative validation approach ensures that each data point is utilized for both training and validation, thus minimizing sample bias and providing a thorough evaluation of the model's effectiveness [29].

The chief advantage of k -fold cross-validation lies in its ability to generate a reliable estimate of the model's performance. By averaging the results from multiple training and validation iterations, this method delivers a more dependable measure of accuracy and generalizability compared to a single train-test split. This approach is particularly beneficial in situations where data is limited or the risk of overfitting is substantial. It allows researchers to fine-tune the model's parameters and confidently select the best-performing model, as it provides a more realistic representation of the

model's performance on unseen data [29].

Furthermore, the structured repetition of training and validation phases helps identify variability in the model's predictions, highlighting its strengths and weaknesses across different subsets of the dataset. This can be crucial for models intended for clinical applications, where prediction reliability is paramount. Figure 2.7 illustrates the configuration of training and validation folds in a 5-fold cross-validation setup and the workflow from training through model evaluation, emphasizing the systematic approach in model testing [29].

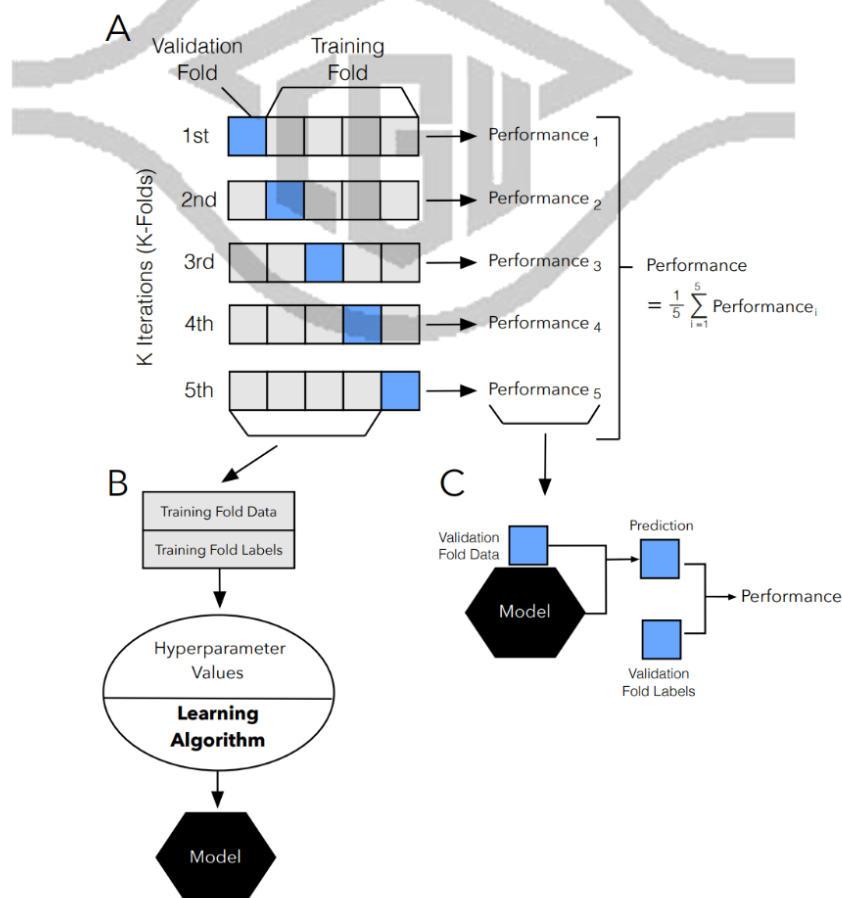


Figure 2.7 K-fold cross-validation [29]

This approach not only reinforces the validation of predictive models but also ensures that the conclusions drawn from the data are more statistically sound, making k-fold cross-validation a cornerstone in developing and assessing predictive models in medical research and other fields. As documented in [29], this method is integral in refining models to achieve higher accuracy and reliability, supporting the advancement of personalized medicine and decision-making processes in healthcare.

2.4.4 Multilayer Perceptron in Medical Imaging

Multilayer Perceptron (MLP) neural networks have demonstrated significant potential in various medical imaging applications, particularly in classification and prediction tasks. The MLP architecture, recognized for its capacity to capture intricate relationships within high-dimensional data, is ideally suited for incorporating radiomic features extracted from medical images. In a study by [7], an MLP-based model was employed to forecast recurrence patterns in glioblastoma using dynamic susceptibility contrast-enhanced MRI. The findings highlighted the effectiveness of merging radiomic features with MLP to enhance predictive accuracy. This methodology enables the modeling of complex patterns within MRI data, essential for identifying recurrence patterns and guiding clinical decision-making.

Similarly, research conducted by [31] investigated the use of MLP networks to differentiate glioblastoma from primary central nervous system lymphoma. By utilizing a comprehensive set of MRI-derived radiomic features, the MLP classifier achieved notable accuracy in distinguishing between these two conditions. The study emphasizes the potential of MLP in improving diagnostic precision and facilitating personalized treatment approaches.

The integration of advanced feature selection techniques, such as Support Vector Machine-Recursive Feature Elimination (SVM-RFE), with MLP networks can further boost their performance. As detailed in [29], the SVM-RFE method systematically reduces the high-dimensional feature space to the most predictive features, enhancing both the model's efficiency and interpretability.

This combination is vital for managing the extensive data generated in radiomic studies and ensuring the clinical relevance of predictive models. The application of MLP networks in conjunction with radiomic features presents a powerful strategy for medical imaging analysis, especially for complex conditions like glioblastoma. By meticulously extracting and selecting the most pertinent features, these models can significantly improve the accuracy of recurrence pattern predictions, ultimately contributing to improved patient outcomes and advancing the field of

precision medicine.

2.4.5 SHAP Values

In the context of interpreting model predictions, SHAP (SHapley Additive exPlanations) values offer a unified and theoretically sound measure of feature importance. Based on the Shapley values from cooperative game theory, SHAP values quantify the marginal contribution of each feature to the prediction outcome.

SHAP values make the interpretation of model predictions more transparent and understandable by quantifying the marginal contribution of each feature to the prediction. In the SHAP framework, the explanation model is a simplified linear model expressed as [32]:

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i$$

where z' represents the simplified input features, and ϕ_i are the SHAP values that quantify the contribution of each feature to the prediction. These values are computed by considering all possible subsets of features and comparing the model predictions with and without the feature in question.

Mathematically, the SHAP value for feature i is given by [32]:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} \left[f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S) \right]$$

where F is the set of all features, S is a subset of F that does not include

i , and f_S represents the model trained on the features in S .

The process of calculating SHAP values involves three main steps:

- **Feature Impact Calculation:** Assess the impact of each feature on the prediction by including and excluding it.
- **Subset Averaging:** Determine the marginal contribution of each feature across all subsets and average these contributions.
- **Weighted Average:** Compute the final SHAP value as a weighted average of contributions across all subsets.

SHAP values offer an intuitive understanding of feature contributions, making model predictions transparent and explainable.

This transparency is particularly important in clinical applications, where understanding influential features builds trust in AI models and aids in clinical decision-making.

Using SHAP values enhances model interpretability and practical application in clinical settings. By supporting precision medicine goals, SHAP values help develop personalized treatment plans based on individual patient features, ultimately improving patient outcomes.

Chapter 3 Materials and Methods

3.1 Research Design

This research aims to develop a prediction model for GBM recurrence patterns based on integrating radiomic features and a neural network, as shown in Figure 3.1. This model integrates whole-brain and multimodal MRI imaging radiomic features to enhance the accuracy and reliability of predictions. The data for this research was sourced from Keelung Chang Gung Memorial Hospital and includes preoperative MRI scans of 106 GBM patients covering T1, T1c, T2, and FLAIR sequences.

To ensure data consistency across patients, all images underwent extensive preprocessing steps using the CaPTk, including N4 bias correction, Z-score normalization, image registration, skull-stripping, and label binarization. These preprocessing steps ensure that the images are uniform in terms of intensity and orientation, facilitating accurate feature extraction and analysis. Following the precise segmentation of brain structures using the FreeSurfer SAMSEG algorithm, the entire brain was segmented to examine various functional areas, such as the ventricles, for subsequent analysis. This segmentation process is crucial for isolating regions of interest that are most relevant to the study, including the tumor core and surrounding edema.

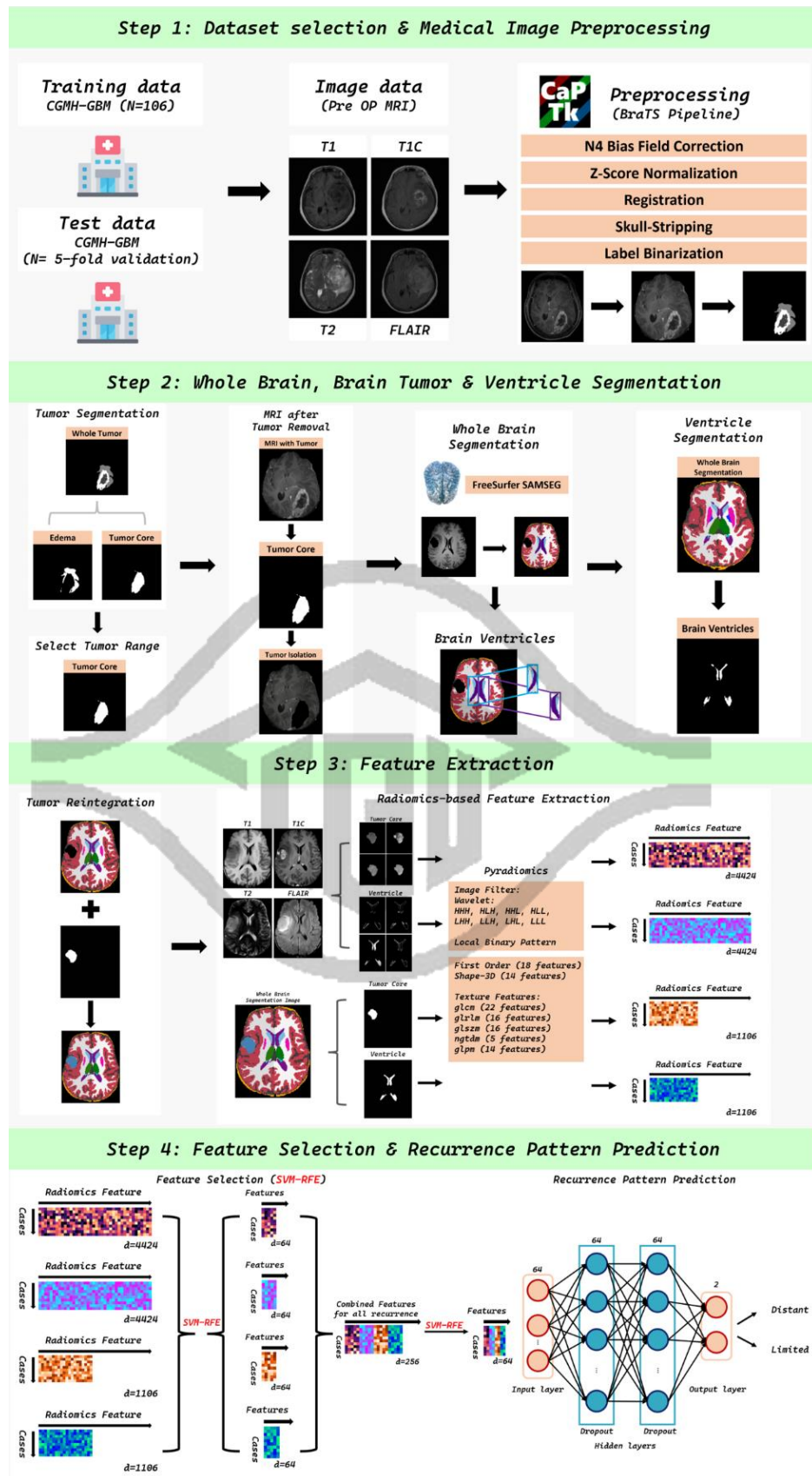


Figure 3.1 Experimental flow chart

By segmenting the whole brain, we can investigate the impact of the tumor on other brain regions, providing a comprehensive understanding of the disease.

Subsequently, the Pyradiomics tool extracted a wide range of radiomic features from both whole-brain and multimodal MRI images. These features include morphological, first-order statistical, and various textural features, such as GLCM, GLRLM, and GLSZM. The SVM-RFE method was utilized to select the most impactful features for the predictive model.

Using the selected features, a deep learning model based on a MLP architecture was trained to predict GBM recurrence patterns, distinguishing between distant and local recurrence. The MLP model includes several hidden layers, each equipped with ReLU activation functions and dropout layers to prevent overfitting. This architecture allows the model to learn complex relationships within the data while maintaining generalizability.

The model was rigorously trained and evaluated using cross-validation methods to ensure robustness and reliability. Five-fold cross-validation was employed to assess the model's performance, providing a comprehensive evaluation across different subsets of the dataset. This method helps to ensure that the model performs consistently and accurately across various patient groups.

This research aims to establish a reliable and precise prediction model to support clinical decision-making, enhance targeted treatment strategies, and improve treatment outcomes and quality of life for GBM patients. By integrating advanced imaging feature extraction and neural network modeling, this study specifically examines the impact of brain ventricle changes on recurrence patterns. The comprehensive analysis of brain ventricles and their relationship with tumor recurrence highlights the potential influence of these regions on predicting disease outcomes more effectively. This approach ultimately contributes to the advancement of precision medicine in oncology by providing insights that could refine treatment plans and improve patient management.

3.2 Experimental Environment

The experimental environment for this study was designed to ensure optimal performance and reliability, combining advanced hardware and specialized software configurations. The primary system used for the experiments was a laboratory computer with specifications detailed in Figure 3.2. This setup included an AMD Ryzen 9 3900X 12-Core Processor, 128 GB of DDR4 memory, an X570S AORUS PRO AX motherboard, an NVIDIA A5000 GPU, and a high-resolution display of 3840x2160, all running on Ubuntu 20.04.6 LTS with a 2 TB SSD for fast

data access and storage.

Operating System	Ubuntu 20.04.6 LTS
CPU	AMD Ryzen 9 3900X 12-Core Processor
Memory	128 GB DDR4
Motherboard	X570S AORUS PRO AX (Gigabyte Technology Co., Ltd.)
Graphics	NVIDIA A5000 GPU
Resolution	3840x2160
Storage	2 TB SSD

Figure 3.2 Experimental flow chart

Software Tools	Version
Programming Language	Python 3.9
Deep Learning Framework	PyTorch 2.2.2
Imaging Processing Tools	
Cancer Imaging Phenomics Toolkit (CaPTk)	1.9.0
FreeSurfer	7.4.1
Radiomics Tools	
pyradiomics	3.0.1
Machine Learning Tools	
scikit-learn	1.1.3
Data Processing and Analysis Tools	
numPy	1.25.2
pandas	1.5.2
Visualization Tools	
Matplotlib	3.6.2
Seaborn	0.13.2
Other Tools	
Simple ITK	2.2.0
SimpleITK-SimpleElastix	2.0.0rc2.dev908
tqdm	4.64.1
six	1.16.0
nibabel	4.0.2

Figure 3.3 Experimental flow chart

The software tools and libraries used in this study are listed in Figure

3.3. This combination ensured the efficiency and accuracy of data processing, model training, and analysis.

This combination of high-performance hardware and robust software tools ensured the efficiency and accuracy of data processing, model training, and analysis throughout the study.

3.3 Data Collection

The data utilized in this study were sourced from Keelung Chang Gung Memorial Hospital (IRB No. 202301763B0) and included preoperative MRI images from 106 GBM patients. All imaging data were rigorously reviewed and confirmed by medical professionals, ensuring the quality and accuracy of the diagnoses. For the analysis of tumor recurrence patterns, these cases were classified into two categories based on the location of recurrence: 60 cases of distant recurrence and 46 cases of limited recurrence. This classification provided a crucial foundation for subsequent feature extraction and model training aimed at accurately predicting and analyzing the behavior of tumor recurrence, as depicted in Figure 3.4.

3.4 Data Preprocessing

During this research preprocessing stage, the initial data was implemented through several meticulous procedures to ensure the

reliability and uniformity of the collected MRI images. As depicted in Figure 3.4, the processing commenced with the CaPTk, which incorporates the BraTS Pipeline. This approach began with applying N4 bias field correction, a pivotal technique to address MRI scan non-uniformities. This step standardized intensity levels across the MRI image dataset, enhancing the comparability and consistency of scans.

All MRI scans underwent Z-score normalization following bias correction, transforming pixel intensity values to a common scale to mitigate variances across different patient scans. This normalization process is essential for data comparability and reliability, particularly in studies involving multiple patients.

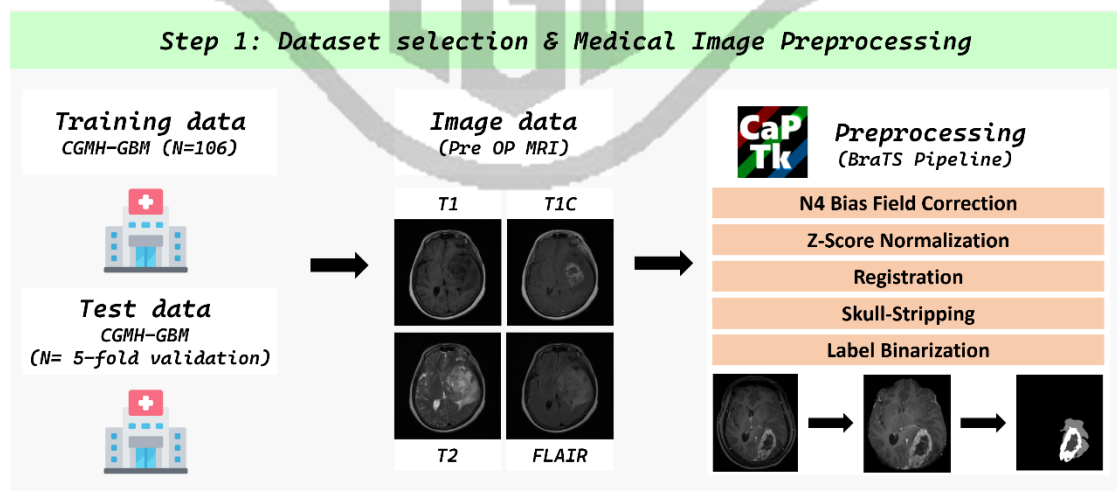


Figure 3.4 Dataset selection & Medical Image Preprocessing

Subsequent spatial registration ensured precise alignment of each sequence with a standard brain atlas, maintaining consistent anatomical positioning across the dataset. This step facilitated accurate feature

extraction and in-depth analysis later in the study. Additionally, all MRI scans were adjusted to a uniform spatial resolution of 240 x 240 x 155.

Skull stripping was then conducted to remove non-brain tissues from the images, sharply focusing the analysis on pertinent brain structures and significantly enhancing the accuracy of subsequent tumor segmentation. Using advanced algorithms, precise delineation of glioblastoma boundaries was performed on T1, T1C, T2, and FLAIR sequences.

The segmentation process culminated with applying label binarization to the segmented tumor regions, isolating these areas for focused analysis in subsequent stages. This preprocessing framework laid a robust foundation for the detailed extraction of radiomic features and the construction of a predictive model for glioblastoma recurrence patterns, which is integral to the aims of this research.

After the initial preprocessing steps, the focus shifted to precisely segmenting brain tumors and their surrounding functional areas, as depicted in Figure 3.5. This segmentation phase was crucial for isolating specific brain structures and tumors, enabling detailed radiomic feature extraction essential for the subsequent predictive modeling of glioblastoma recurrence patterns.

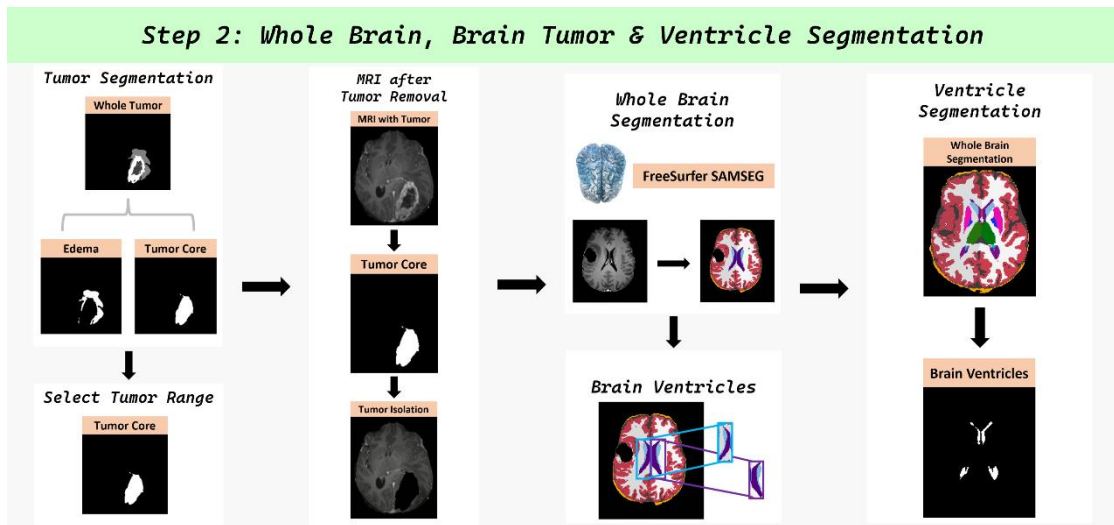


Figure 3.5 Whole Brain, Brain Tumor & Ventricle Segmentation

The segmentation process began by delineating the entire tumor from the MRI scans, followed by a more focused segmentation of the tumor core, which typically contains the most biologically aggressive parts of the tumor. Clinical neurosurgeons rely on the differentiation between edematous tissue and the tumor core to evaluate patient conditions, as the definition of edema can vary between individuals. Focusing on the tumor core provides more precise clinical assessment and is critical for evaluating potential recurrence patterns.

Once the tumor core was isolated, MRI scans depicting the brain post-tumor removal were analyzed to ensure no residual tumor tissue remained, providing a baseline for assessing the efficacy of surgical interventions and for monitoring potential recurrence. Furthermore, the entire brain of each case was segmented using FreeSurfer's SAMSEG algorithm, providing a

comprehensive map that highlighted all functional brain regions. This detailed depiction allowed clinicians to precisely observe the spatial relationships between the tumor and critical brain structures, which is vital for assessing the tumor's impact on overall brain function and for planning further therapeutic measures.

Additionally, the segmentation of the brain ventricles was performed, capturing their location and morphology which are critical for understanding changes in cerebrospinal fluid dynamics and potential intracranial pressure alterations caused by tumor growth. The segmented ventricles were used as masks in subsequent analyses to evaluate deformations or displacements caused by the tumor, providing insights that are crucial for predicting complications associated with tumor expansion and for planning effective treatment strategies. This detailed segmentation approach not only facilitates the extraction of relevant radiomic features but also enhances the anatomical and physiological basis for developing a robust model capable of predicting glioblastoma recurrence patterns.

3.5 Feature Extraction

In this study, extensive radiomics feature extraction was performed on data from four different MRI imaging modalities (T1, T1c, T2, FLAIR), aiming to achieve an in-depth analysis of glioma (GBM) and its interaction

with related ventricular structures from multiple perspectives.

The feature extraction process included not only the primary MRI images but also integrated whole-brain segmentation images, enhancing the identification accuracy of disease-specific features.

A critical aspect of this integration was the 'Tumor Reintegration Process', wherein tumor core masks were precisely aligned with the global brain architecture provided by FreeSurfer's SAMSEG algorithm, as detailed in Figure 3.6.

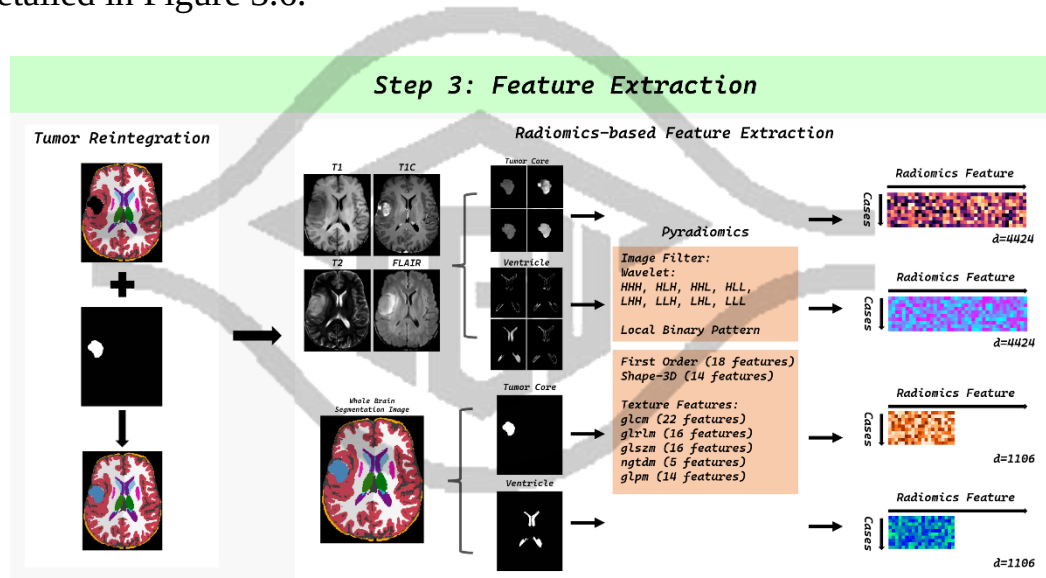


Figure 3.6 Feature Extraction

This reintegration step was crucial for merging localized tumor features with broader brain features, ensuring that spatial correlations and interactions between tumor growth and brain structures were accurately captured. The enhanced specificity of the radiomic features extracted, particularly in distinguishing between tumor and adjacent brain tissues, is

pivotal for accurately modeling the progression and potential recurrence of glioblastoma, thereby improving the predictive power of the study.

Following the meticulous segmentation processes as described and illustrated in Figure 3.6, the study advances into the core of its analytical phase—feature extraction. This segment of the research harnesses advanced radiomic techniques to transform raw MRI data into a rich dataset ripe for computational analysis. The extraction of radiomic features is executed across four key areas, each tailored to unveil unique aspects of glioblastoma and its environment within the brain.

This detailed extraction process allows for the comprehensive mapping of tumor characteristics and their potential impact on brain functionality, crucial for predicting glioblastoma recurrence patterns:

1. Feature extraction of brain tumors from MRI images:

These included 18 first-order features, 14 shape features in 3D, 22 GLCM features, 16 GLSZM features, 16 GLRLM features, 5 NGTDM features, and 14 GLDM features. To enhance the dimensionality and improve the analytical accuracy of the image data, wavelet transformation and Local Binary Pattern (LBP) filtering were additionally employed for feature extraction. The wavelet transformation produced eight decomposed images labeled HHH, HLH, HHL, HLL, LHH, LLH, LHL, and

LLL, where 'H' denotes a high-pass filter and 'L' indicates a low-pass filter. The derived wavelet images were then processed to extract first-order and grayscale variation features, including GLCM, GLRLM, GLSZM, NGTDM, and GLDM. This led to the generation of 728 wavelet features by applying 18 first-order features and 73 grayscale variation features across the eight decomposed images. Similarly, LBP were applied to the original MRI sequences, extracting 18 first-order features and 73 grayscale variation features across three different LBP images, resulting in 273 LBP features. Finally, 1106 radiomics features were extracted from the four MRI modalities—T1, T1c, T2, and FLAIR and this culminated in a comprehensive dataset of 4424 features per patient.

2. Feature Extraction from MRI Images of the Ventricular

System: For each MRI modality (T1, T1c, T2, FLAIR), 105 fundamental radiomics features were extracted from the ventricular structures, encompassing first-order, shape, and texture features. Further feature extraction was performed using wavelet transformation and LBP techniques, generating 728 wavelet features and 273 LBP features, totaling 4424 features.

3. Feature Extraction from Whole Brain Segmented Images for

Brain Tumors: For the images processed through FreeSurfer's

whole-brain segmentation, this study applied the same feature extraction techniques used for the original MRI images. A total of 1106 features were extracted from the brain tumor regions, which included 105 basic features, 728 features enhanced through Wavelet transformation, and 273 features obtained using LBP.

4. **Feature Extraction from Whole Brain Segmented Images for the Ventricular System:** Similarly, for the whole-brain segmented images processed by FreeSurfer, a parallel set of features was extracted from the ventricular structures. This part also resulted in 1106 features extracted from the ventricles, comprising 105 basic features, 728 via Wavelet transformation, and 273 LBP features.

These feature extraction steps provide comprehensive radiomic insights into the pathological characteristics of brain tumors and ventricular systems, facilitating advanced research and diagnostic capabilities in neurology.

3.6 Feature Selection

We focus on identifying the most predictive features from the extensive set of features extracted in Section 3.6 from various MRI and whole brain images. Selecting the most informative features is crucial for

constructing a streamlined and effective predictive model, which enhances the model's predictive performance and improves computational efficiency and interpretability.

Initially, for each category of tumor and ventricular images, the feature set included basic features from the original MRI sequences and high-dimensional features enhanced through wavelet and LBP transformations. However, considering computational efficiency and model interpretability, not all features were included in the final predictive model. Thus, we employed the SVM-RFE method to identify and select the top 64 most important features for each image type.

Enhancing computational efficiency and model interpretability is particularly vital in clinical settings, where timely and accurate predictions are crucial. By narrowing down the feature set, the model's computational demands are significantly reduced, enabling faster processing times which are essential for real-time decision-making in a clinical environment. Furthermore, a more interpretable model allows clinicians to better understand the factors driving the predictions, thereby increasing trust and facilitating adoption in clinical workflows.

In the SVM-RFE approach, we first established an SVM classifier with a linear kernel as the base learner to handle the feature space's high dimensionality effectively. The SVM-RFE process involves training the

model on all features and then using the classifier's weight coefficients to score and rank the features based on their contribution to the model's decision boundary. The least contributing features are recursively eliminated until only the predetermined number of features remains. These selected features are considered to have the highest impact on predicting the recurrence patterns of GBM.

This methodology leverages the theoretical strengths of Support Vector Machines, particularly their capacity to handle high-dimensional data spaces effectively. This method is especially suitable for radiomics, where the feature space is often vast and complex. Studies have shown that SVM-RFE can effectively reduce the feature space while maintaining, or even enhancing, the model's predictive power by focusing on the most relevant features. This makes SVM-RFE a robust choice for feature selection in medical imaging applications.

During the feature selection phase, we meticulously filtered features for each type of tumor and ventricular image from the original, wavelet-transformed, and LBP-transformed MRI images. Ultimately, this selection process from the extensive dataset resulted in a final set of features. We combined the selected features from the various image types and applied SVM-RFE once more to identify the top 64 features from this integrated set, resulting in a more streamlined and powerful feature set for training the

predictive models.

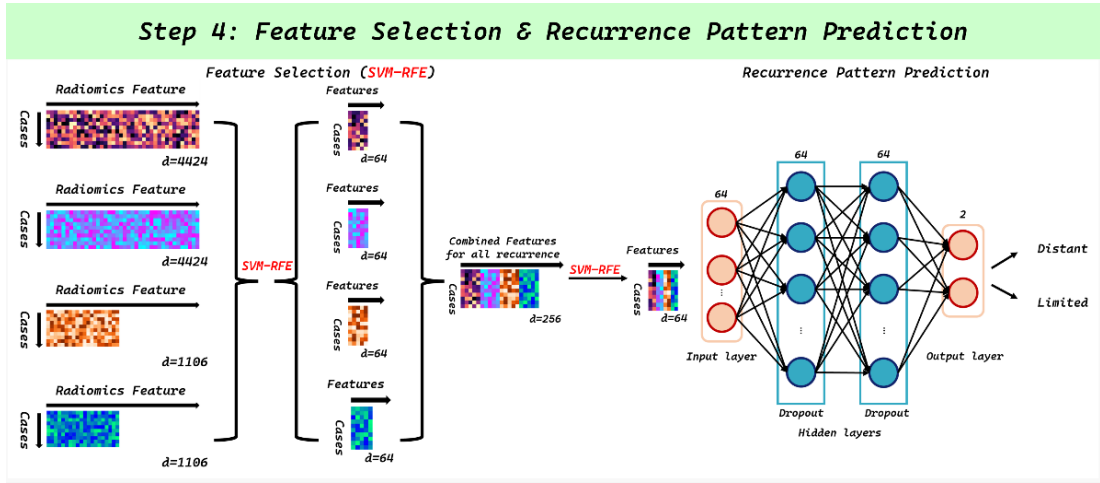


Figure 3.7 Feature Selection & Recurrence Pattern Prediction

To illustrate, Figure 3.7 shows the steps involved in feature selection using SVM-RFE, where the radiomics features are filtered, ranked, and the top features are selected for the final model.

We also experimented with different numbers of features, selecting sets of 8, 16, 32, 64, 128, 256, 512, 1024, and eventually all available features. This approach allowed us to assess the impact of the number of features on the model's performance and computational efficiency. By evaluating different feature set sizes, we ensured that the model retained sufficient information to accurately predict glioblastoma recurrence patterns while maintaining manageable computational requirements.

This feature selection process optimized the model and enhanced the interpretability of the results. By choosing features with high predictive value, we reduced the complexity of the model while retaining sufficient

information to predict the recurrence patterns of glioblastoma accurately. The selected features also aid in understanding the biological processes underlying disease progression, which will be further discussed and validated in subsequent chapters.

3.7 Model Structure

The model structure for predicting glioblastoma (GBM) recurrence patterns integrates radiomic features with a neural network, specifically using a Multi-Layer Perceptron (MLP) architecture. This approach leverages high-dimensional data derived from MRI scans and advanced image processing techniques to accurately predict recurrence patterns, as depicted in Figure 3.8.

Two different models, Model A and Model B, were constructed and evaluated to compare their performance:

- **Model A:** The MLP architecture for Model A is designed with smaller hidden layers, making it computationally efficient and less prone to overfitting.
- **Input Layer:** Receives 64 combined features selected through the SVM-RFE process. These features include morphological, first-order statistical, and various textural characteristics extracted from whole-brain and multimodal MRI images.

- Hidden Layers:
 - First Hidden Layer: Consists of 64 neurons. It applies ReLU activation functions to introduce non-linearity, allowing the network to learn complex patterns. Batch normalization is used to stabilize and accelerate the training process.
 - Second Hidden Layer: Another layer with 64 neurons, also utilizing ReLU activation and batch normalization. This layer further processes the data, enabling the model to refine its understanding of the input features.
 - Dropout Layers: Dropout is applied between the hidden layers to prevent overfitting. By randomly setting a fraction of the input units to zero at each update during training, dropout helps the model generalize better.
- Output Layer: Consists of 2 neurons, representing the two classes of recurrence patterns (distant and limited). A softmax activation function is applied to generate probabilities for each class, allowing the model to make a definitive classification.
- Model B: The MLP architecture for Model B is designed with larger hidden layers, capable of capturing more complex patterns.
- Input Layer: Receives 64 combined features selected through

the SVM-RFE process.

- Hidden Layers:
 - First Hidden Layer: Consists of 1024 neurons with ReLU activation functions and batch normalization.
 - Second Hidden Layer: Another layer with 1024 neurons, also utilizing ReLU activation and batch normalization.
 - Dropout Layers: Dropout is applied between the hidden layers to prevent overfitting.
- Output Layer: Consists of 2 neurons, representing the two classes of recurrence patterns (distant and limited), with a softmax activation function.

Using the selected features, both Model A and Model B were trained specifically to predict GBM recurrence patterns, distinguishing between distant and local recurrence. Each model includes several hidden layers and dropout layers to prevent overfitting and was rigorously trained and evaluated using cross-validation methods to ensure generalizability and reproducibility across different patient datasets.

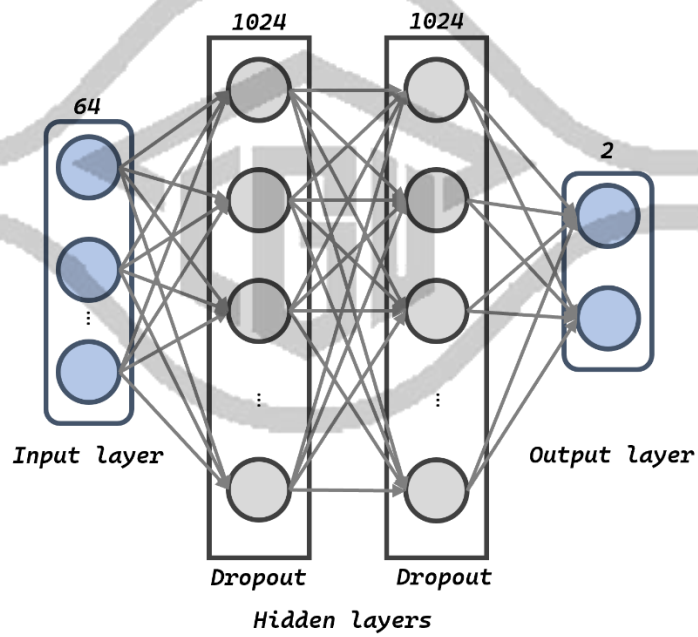
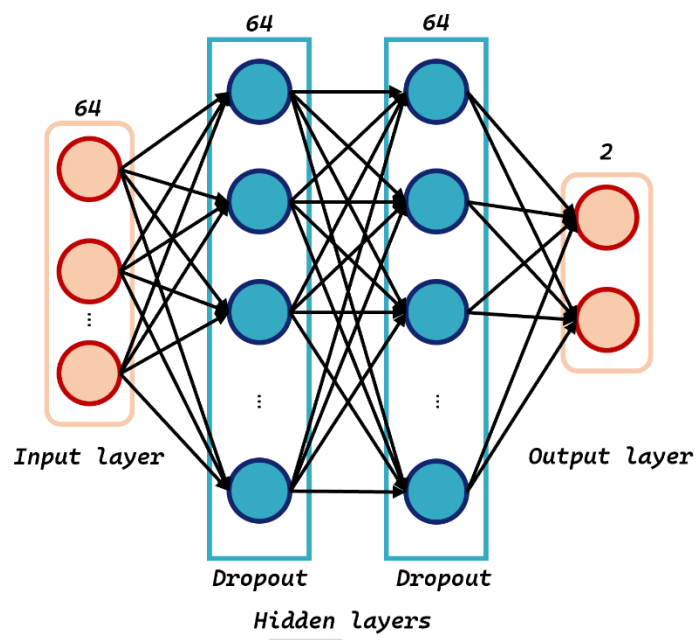


Figure 3.8 Model A & Model B Structure

Chapter 4 Experimental Results

In this chapter, we present the experimental results obtained from applying the developed neural network models, Model A and Model B, to predict GBM recurrence patterns using radiomic features extracted from multimodal MRI images. The results are analyzed to evaluate the models' performance, focusing on accuracy, reliability, and potential clinical implications of the predictions. The experimental results are organized into several sections, each detailing the performance of the models under different conditions and configurations, providing a comprehensive understanding of their capabilities and limitations.

4.1 Performance Metrics

This section details the performance metrics used to evaluate our neural network model for predicting glioblastoma (GBM) recurrence patterns. Our dataset consists of MRI features extracted from 106 patients with recurrent GBM. The classification task focuses on distinguishing between two types of recurrence patterns: Distant recurrence and Limited recurrence. To comprehensively assess the model's performance, we utilized several key metrics: Accuracy, Average Accuracy, Standard Deviation (std), and the Confusion Matrix.

4.1.1 5-Fold Cross-Validation

In this study, we employed a five-fold cross-validation method to evaluate the performance of our neural network model for predicting GBM recurrence patterns. This method is particularly effective for small datasets and provides a robust estimate of model performance by ensuring that each data point is used for both training and validation. Our dataset consists of MRI features extracted from 106 patients with recurrent GBM. The classification task focuses on distinguishing between two types of recurrence patterns: Distant recurrence and Limited recurrence.

Processing:

- **Data Division:** The entire dataset is randomly divided into five equal parts, or "folds". Each fold contains approximately 21 patients.
- **Training and Validation:** For each iteration, four folds are combined to create the training set, while the remaining fold serves as the validation set. This process is repeated five times, ensuring each fold is used as the validation set exactly once..
- **Model Training:** During each iteration, the model is trained on the training set using the selected features. The training process involves optimizing model parameters and mitigating potential overfitting.

- **Model Validation:** The trained model is then validated on the validation set, and performance metrics such as accuracy, average accuracy, and standard deviation are computed..
- **Aggregation of Results:** Upon completing all five iterations, the results from each fold are aggregated to provide a comprehensive assessment of the model's performance. This includes calculating the mean and standard deviation of accuracy across all folds..

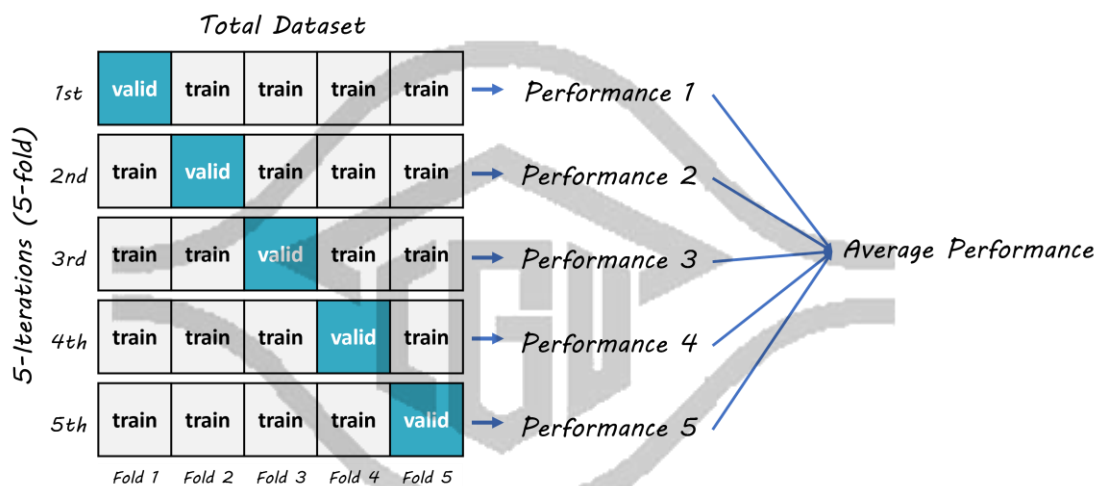


Figure 4.1 5-Fold Cross-Validation Structure

The inclusion of Figure 4.1 provides a visual representation of the five-fold cross-validation process, enhancing the reader's understanding of how the dataset was divided and used for model training and validation. It also helps identify potential overfitting issues and ensures that the model generalizes well to new, unseen data. By applying this rigorous validation method, we ensure the model's performance metrics are reliable and reflect its true predictive capabilities.

4.1.2 Accuracy & Confusion Matrix

Accuracy measures the proportion of correctly classified instances among the total instances evaluated. It is calculated as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- TP (True Positive) represents the number of correctly predicted Limited recurrences.
- TN (True Negative) represents the number of correctly predicted Distant recurrences.

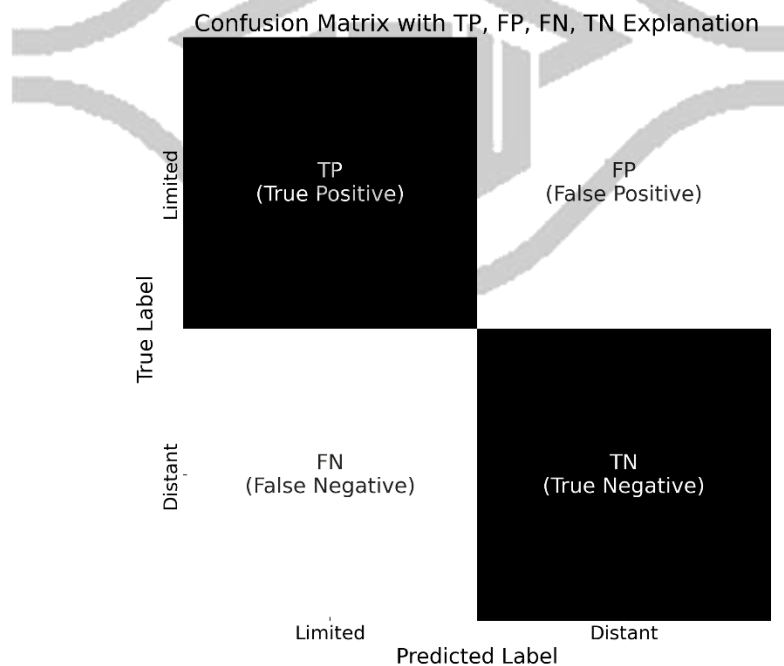


Figure 4.2 Confusion Matrix

- FP (False Positive) indicates the number of Distant recurrences incorrectly predicted as Limited.

- FN (False Negative) indicates the number of Limited recurrences incorrectly predicted as Distant.

The confusion matrix, illustrated in Figure 4.2, visually represents these metrics, providing a clear view of the model's performance across different prediction categories. It is a valuable tool for evaluating the classification performance of the model. It helps identify how many instances were correctly or incorrectly classified, which is crucial for understanding the model's accuracy and areas for improvement.

4.1.3 Average Accuracy

Average Accuracy is a critical metric for evaluating the overall performance of our neural network model across multiple validation folds. This metric provides a robust measure of the model's consistency and reliability in predicting GBM recurrence patterns by averaging the accuracy obtained from each fold during the five-fold cross-validation process.

The average accuracy is calculated by taking the mean of the accuracy values obtained from each fold in the five-fold cross-validation. The formula for average accuracy is:

$$\text{Average Accuracy} = \frac{1}{5} \sum_{i=1}^5 \text{Accuracy}_i$$

where:

- 5 is the number of folds in the cross-validation.
- $Accuracy_i$ is the accuracy obtained in the i fold.

This process ensures that the average accuracy reflects the model's performance across different subsets of the data, mitigating the impact of any particular fold that may have atypical data characteristics.

For our study involving 106 patients with recurrent GBM, average accuracy is crucial for several reasons:

- **Consistency:** By evaluating the model's performance across multiple folds, average accuracy provides a more reliable measure of its true predictive capability. This is particularly important in medical applications where consistent performance across different patient subsets is essential.
- **Robustness:** Averaging the accuracy over multiple folds reduces the likelihood that the model's performance is influenced by the specific partitioning of the data. This robustness ensures that the model is not overfitting to particular folds but generalizes well to unseen data.
- **Clinical Relevance:** In predicting GBM recurrence patterns, high average accuracy indicates that the model can reliably differentiate between distant and limited recurrence types, which is critical for

informing treatment decisions and improving patient outcomes.

In our study, we performed five-fold cross-validation, meaning the dataset was divided into five equal parts, and the model was trained and validated five times, each time using a different fold as the validation set and the remaining folds as the training set. The average accuracy was then calculated by averaging the accuracy scores from these five validation rounds. The high average accuracy achieved in our experiments demonstrates the effectiveness of our model in predicting GBM recurrence patterns. It provides confidence that the model performs well not only on the training data but also on new, unseen patient data, which is crucial for its application in a clinical setting.

4.1.4 Standard Deviation

Standard Deviation is a vital metric for evaluating the variability and reliability of our neural network model's performance in predicting GBM recurrence patterns. This metric quantifies the dispersion of accuracy scores across the different folds in the five-fold cross-validation process, providing insights into the model's consistency.

The standard deviation of the accuracy scores is calculated using the following formula:

$$\text{Standard Deviation} = \sqrt{\frac{1}{5} \sum_{i=1}^5 (\text{Accuracy}_i - \text{Average Accuracy})^2}$$

where:

- 5 is the number of folds in the cross-validation.
- Accuracy_i is the accuracy obtained in the i fold.
- Average Accuracy is the mean accuracy across all five folds.

This formula measures the spread of accuracy scores around the average accuracy, indicating how much the accuracy varies from fold to fold.

In the context of our study, involving 106 patients with recurrent GBM, the standard deviation of accuracy is important for several reasons:

- **Consistency:** A low standard deviation indicates that the model's accuracy is consistent across different folds, suggesting that the model generalizes well and is not overly sensitive to the specific training and validation split.
- **Reliability:** A high standard deviation would suggest variability in the model's performance, indicating potential issues with overfitting or sensitivity to particular subsets of the data. This could undermine the model's reliability in clinical applications.
- **Confidence in Predictions:** In medical contexts, especially when predicting GBM recurrence patterns, it is crucial to have a model

that performs reliably across different patient subsets. A low standard deviation reinforces the confidence in the model's predictions, making it more suitable for clinical decision-making.

In our five-fold cross-validation study, the standard deviation was calculated based on the accuracy scores from each fold. This process involved:

- **Calculating Accuracy for Each Fold:** The accuracy for each of the five folds was computed individually.
- **Determining Average Accuracy:** The average accuracy was calculated by averaging the accuracy scores from all five folds.
- **Computing Standard Deviation:** The standard deviation was then computed to measure the variation of these accuracy scores from the average accuracy.

4.2 Results

4.2.1 Feature Combination

To enhance the predictive power of our model, we evaluated different feature sets, each consisting of 64 features:

- **MRI Brain Tumor:** 64 features from MRI images of the brain tumor region.
- **MRI Ventricle:** 64 features from MRI images of the brain ventricle

region.

- Whole Brain Image Brain Tumor: 64 features from whole brain images focusing on the brain tumor region.
- Whole Brain Image Brain Ventricle: 64 features from whole brain images focusing on the brain ventricle region.
- MRI (Brain Tumor + Ventricle): 64 features selected from both MRI brain tumor and ventricle regions.
- Whole Brain Image (Brain Tumor + Ventricle): 64 features were selected from both whole brain images, focusing on the brain tumor and ventricle regions.
- MRI + Whole Brain Image (Brain Tumor + Ventricle): 64 features selected from a combination of MRI and whole brain images focusing on both brain tumor and ventricle regions.

By evaluating these different feature sets, we aim to determine the optimal combination that maximizes the predictive accuracy of our model.

4.2.2 Model Evaluation and Results (64 Features)

This section details the classification accuracy performance of different feature combinations. Each feature set consists of 64 features and was evaluated using five-fold cross-validation.

Table 1.1 64 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor	100	95.24	90.48	95.24	100	96.19 ± 3.56
MRI Ventricle	100	100	90.48	100	100	98.10 ± 3.81
Whole Brain Image Brain Tumor	77.27	80.95	52.38	66.67	66.67	68.79 ± 9.98
Whole Brain Image Brain Ventricle	81.82	85.71	57.14	76.19	76.19	75.41 ± 9.82
MRI Tumor+Ventricle	95.45	95.24	90.48	95.24	100	95.28 ± 3.01
Whole Brain Image Tumor+Ventricle	86.36	76.19	71.43	95.24	85.71	82.99 ± 8.35
MRI + Whole Brain Image Tumor+Ventricle	100	100	85.71	100	100	97.14 ± 5.72

Table 1.1 presents the specific accuracies and average accuracies for these feature combinations across the five-fold cross-validation. The results indicate that the combined features of MRI brain tumor and ventricle regions achieve higher accuracy, demonstrating the effectiveness of feature combination in improving model classification accuracy.

Evaluating the model with different combinations of 64 features, it was found that combining brain tumor and ventricle features significantly enhances the classification accuracy, especially for the MRI image feature sets. Therefore, subsequent experimental results will focus on using MRI image feature combinations to further explore the impact of different feature sets on model performance.

4.2.3 8 Features

In this section, we evaluate the predictive power of the models based on accuracy and feature importance. The most important features from each model are selected for detailed explanation. Table 1.2 shows the model accuracy results for different feature combinations.

Table 1.2 8 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	81.82	71.43	71.43	80.95	95.24	80.17 $\pm$ 8.75
MRI Brain Tumor (Model B)	68.18	71.43	71.43	80.95	85.71	77.54 $\pm$ 9.04
MRI Ventricle (Model A)	77.27	80.95	61.90	85.71	80.95	77.36 $\pm$ 8.18
MRI Ventricle (Model B)	77.27	80.95	66.67	95.24	80.95	80.22 $\pm$ 9.15
MRI Tumor+Ventricle (Model A)	86.36	76.19	71.43	76.19	61.90	74.41 $\pm$ 7.93
MRI Tumor+Ventricle (Model B)	63.64	76.19	71.43	85.71	61.90	71.77 $\pm$ 8.69

For brain tumor features, the most important feature for Model A is *t1-tumor-wavelet-LLH_ngtdm_Complexity*. This feature measures the complexity of wavelets in T1-weighted tumor images. High complexity indicates more detail and variability in the images, which is crucial for predicting tumor recurrence patterns. Correspondingly, the most important

feature for Model B is *t1-tumor-wavelet-LHH_gldm_DependenceEntropy*. This feature reflects the dependence entropy in T1-weighted tumor images, where higher dependence entropy indicates more complex relationships in the image, aiding the model in accurately capturing tumor characteristics and recurrence patterns.

Table 1.3 MRI Brain Tumor (Model A) 8 Features

Feature	Importance
t1-tumor-wavelet-LLH_ngtdm_Complexity	0.975361368
t1-tumor-wavelet-LHH_gldm_DependenceEntropy	0.952692929
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	0.90128246
flair-tumor-lbp-3D-k_glrIm_ShortRunLowGrayLevelEmphasis	0.560825347
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	0.557291374
flair-tumor-wavelet-HLL_firstorder_Kurtosis	0.553812958
flair-tumor-wavelet-HLH_firstorder_Minimum	0.429236182
t1-tumor-wavelet-LHL_ngtdm_Busyness	0.284084885

Table 1.4 MRI Brain Tumor (Model B) 8 Features

Feature	Importance
t1-tumor-wavelet-LHH_gldm_DependenceEntropy	2.362686997
t1-tumor-wavelet-LLH_ngtdm_Complexity	2.09290116
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	1.89099014
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	1.447912777
flair-tumor-lbp-3D-k_glrIm_ShortRunLowGrayLevelEmphasis	1.344370597
flair-tumor-wavelet-HLH_firstorder_Minimum	1.083381211
flair-tumor-wavelet-HLL_firstorder_Kurtosis	0.955018694
t1-tumor-wavelet-LHL_ngtdm_Busyness	0.510960342

For ventricle features, the most important feature for Model A is *t1-ventricle-wavelet-HHH_glcm_InverseVariance*. This feature measures the inverse variance in T1-weighted ventricle images. Inverse variance indicates variability in the gray level values of the images, which is important for analyzing ventricle morphology changes and their impact on

the tumor. In Model B, the most important feature is also *t1-ventricle-wavelet-HHH_glcmlInverseVariance*. High inverse variance in the ventricle images helps the model better understand ventricle changes affecting tumor recurrence.

Table 1.5 MRI Ventricle (Model A) 8 Features

Feature	Importance
t1-ventricle-wavelet-HHH_glcmlInverseVariance	0.76891623
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	0.67519815
flair-ventricle-lbp-3D-m2_firstorder_Range	0.633656689
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	0.561353323
flair-ventricle-lbp-3D-m1_firstorder_Skewness	0.472361019
t1ce-ventricle-original_gldm_DependenceNonUniformity	0.402782143
t1-ventricle-lbp-3D-m2_glszm_LargeAreaEmphasis	0.371108221
t1-ventricle-lbp-3D-m1_firstorder_Range	0.133965656

Table 1.6 MRI Ventricle (Model B) 8 Features

Feature	Importance
t1-ventricle-wavelet-HHH_glcmlInverseVariance	1.688760581
flair-ventricle-lbp-3D-m2_firstorder_Range	1.598317164
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	1.340543561
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	1.252788143
flair-ventricle-lbp-3D-m1_firstorder_Skewness	1.100964742
t1ce-ventricle-original_gldm_DependenceNonUniformity	0.706793937
t1-ventricle-lbp-3D-m2_glszm_LargeAreaEmphasis	0.657542843
t1-ventricle-lbp-3D-m1_firstorder_Range	0.296157045

For combined brain tumor and ventricle features, the most important feature for Model A is *t1-tumor-wavelet-LHH_gldm_DependenceEntropy*. This feature combines dependence entropy from both brain tumor and ventricle images, reflecting complex spatial relationships crucial for capturing the interactions between the tumor and ventricles.

In Model B, the most important feature is also *t1-tumor-wavelet-*

LHH_gldm_DependenceEntropy, indicating this feature's greater contribution to predictions in a more complex neural network model.

Table 1.7 MRI Tumor + Ventricle (Model A) 8 Features

Feature	Importance
t1-tumor-wavelet-LHH_gldm_DependenceEntropy	0.886497261
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	0.77690287
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	0.624661337
flair-ventricle-lbp-3D-m1_firstorder_Skewness	0.51587926
flair-tumor-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	0.428682388
flair-tumor-wavelet-HLL_firstorder_Kurtosis	0.361843311
t1-tumor-wavelet-LHL_ngtdm_Busyness	0.187142995
t1-ventricle-lbp-3D-m1_firstorder_Range	0.074970712

Table 1.8 MRI Tumor + Ventricle (Model B) 8 Features

Feature	Importance
t1-tumor-wavelet-LHH_gldm_DependenceEntropy	2.256196612
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	1.645631726
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	1.637940716
flair-ventricle-lbp-3D-m1_firstorder_Skewness	1.394403079
flair-tumor-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	1.197690329
flair-tumor-wavelet-HLL_firstorder_Kurtosis	0.904209757
t1-tumor-wavelet-LHL_ngtdm_Busyness	0.585007568
t1-ventricle-lbp-3D-m1_firstorder_Range	0.164109027

Tables 1.3 to 1.8 show the feature importance for different feature combinations. These tables detail the importance values for each feature, helping us understand the predictive capabilities of each model under various feature combinations.

4.2.4 16 Features

In this section, we evaluate the predictive power of the models based on accuracy and feature importance. Table 1.9 shows the model accuracy

results for different feature combinations.

Table 1.9 16 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	90.91	90.48	90.48	85.71	85.71	88.66 ± 2.41
MRI Brain Tumor (Model B)	90.91	90.48	90.48	85.71	85.71	88.66 ± 2.41
MRI Ventricle (Model A)	100	85.71	80.95	95.24	95.24	91.43 ± 7
MRI Ventricle (Model B)	95.45	85.71	80.95	95.24	90.48	89.57 ± 5.6
MRI Tumor+Ventricle (Model A)	90.91	95.24	85.71	100	95.24	93.42 ± 4.81
MRI Tumor+Ventricle (Model B)	90.91	90.48	85.71	95.24	95.24	91.52 ± 3.55

According to Table 1.9, the average accuracy for Model A using brain tumor features is 88.66%, while Model B also has an average accuracy of 88.66%. This indicates that both models perform similarly when using brain tumor features for prediction. For ventricle features, the average accuracy of Model A is 91.43%, and Model B is 89.57%. When combining brain tumor and ventricle features, the average accuracy for Model A is 93.42%, and for Model B, it is 91.52%.

Regarding feature importance, Tables 1.10 and 1.11 show the importance rankings for brain tumor features. For Model A, the most

important feature is *t1-tumor-wavelet-LHH_gldm_DependenceEntropy*.

This feature measures the dependence entropy in T1-weighted tumor images, indicating more detailed and complex relationships within the image, which helps capture tumor characteristics and recurrence patterns. For Model B, the most important feature is also *t1-tumor-wavelet-LHH_gldm_DependenceEntropy*, further confirming the high importance of this feature in different models.

Table 1.10 MRI Brain Tumor (Model A) 16 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-LHH_gldm_DependenceEntropy	1.321305199
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	1.253314097
t1-tumor-wavelet-LLH_ngtdm_Complexity	1.224716199
t2-tumor-wavelet-LLL_glcm_Imc2	0.931548809
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	0.869760587
t1-tumor-wavelet-LLL_glcm_ClusterProminence	0.830255908
flair-tumor-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	0.71038283
flair-tumor-wavelet-HLL_firstorder_Kurtosis	0.688329737
t1ce-tumor-wavelet-LLH_glcm_Imc2	0.632396768
t1ce-tumor-wavelet-HLL_glcm_ClusterShade	0.502298968

Table 1.11 MRI Brain Tumor (Model B) 16 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-LHH_gldm_DependenceEntropy	1.99470021
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	1.73527423
t1-tumor-wavelet-LLH_ngtdm_Complexity	1.583760663
t2-tumor-wavelet-LLL_glcm_Imc2	1.355981382
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	1.226809406
t1-tumor-wavelet-LLL_glcm_ClusterProminence	1.150574956
flair-tumor-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	1.01887576
t1-tumor-wavelet-HHL_glcm_Imc1	0.98815711
flair-tumor-wavelet-HLL_firstorder_Kurtosis	0.933194915
t1ce-tumor-wavelet-LLH_glcm_Imc2	0.92180302

For ventricle features, Tables 1.12 and 1.13 show the important

features of Models A and B. For Model A, the most important feature is *t2-ventricle-wavelet-LLH_glcm_InverseVariance*. This feature measures the inverse variance in T2-weighted ventricle images, indicating the variability of gray levels, which is crucial for analyzing morphological changes in the ventricles and their impact on tumor predictions. For Model B, the most important feature is also *t2-ventricle-wavelet-LLH_glcm_InverseVariance*, highlighting the consistently high importance of this feature across different models.

Table 1.12 MRI Ventricle (Model A) 16 Features (Top 10)

Feature	Importance
t2-ventricle-wavelet-LLH_glcm_InverseVariance	1.248514419
t1-ventricle-wavelet-HHH_glcm_InverseVariance	1.241739733
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	1.092449177
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	1.034079525
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	1.013757688
t1-ventricle-lbp-3D-m2_glszm_LargeAreaEmphasis	0.957708823
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	0.916064951
flair-ventricle-lbp-3D-m1_firstorder_Skewness	0.883197049
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.860955633
t1ce-ventricle-original_gldm_DependenceNonUniformity	0.721306245

Table 1.13 MRI Ventricle (Model B) 16 Features (Top 10)

Feature	Importance
t2-ventricle-wavelet-LLH_glcm_InverseVariance	1.713103126
t1-ventricle-wavelet-HHH_glcm_InverseVariance	1.605180065
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	1.46713617
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	1.43142602
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	1.412678903
t1-ventricle-lbp-3D-m2_glszm_LargeAreaEmphasis	1.38301354
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	1.369899477
flair-ventricle-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	1.13804024
flair-ventricle-lbp-3D-m1_firstorder_Skewness	1.119525237
flair-ventricle-lbp-3D-m2_firstorder_Maximum	1.06268363

Tables 1.14 and 1.15 show the feature importance rankings for combined brain tumor and ventricle features. For Model A, the most important feature is *t1-tumor-wavelet-LLH_ngtdm_Complexity*. This feature combines the complexity in the brain tumor images, reflecting the significant interaction between the tumor and ventricles.

In Model B, the most important feature is *t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis*, indicating that different models rely on different features when combining features.

Table 1.14 MRI Tumor + Ventricle (Model A) 16 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-LLH_ngtdm_Complexity	1.246353964
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	1.235158632
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	1.172430458
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	1.160101349
t1-ventricle-wavelet-HHH_gldm_InverseVariance	1.083464911
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	0.994810272
t2-tumor-wavelet-LLL_gldm_Imc2	0.900798666
flair-ventricle-lbp-3D-m2_firstorder_Range	0.778492087
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	0.688019312
t1ce-tumor-wavelet-LLH_gldm_Imc2	0.679315041

Table 1.15 MRI Tumor + Ventricle (Model B) 16 Features (Top 10)

Feature	Importance
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	1.848892423
t1-ventricle-wavelet-HHH_gldm_InverseVariance	1.783566017
t1-tumor-wavelet-LLH_ngtdm_Complexity	1.717352308
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	1.706638484
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	1.676089043
flair-tumor-wavelet-LLH_glszm_GrayLevelNonUniformity	1.674900571
t2-tumor-wavelet-LLL_gldm_Imc2	1.291218565
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	1.112853548
t1ce-tumor-wavelet-LLH_gldm_Imc2	1.024085522
flair-ventricle-lbp-3D-m2_firstorder_Range	0.978596358

Tables 1.10 to 1.15 show the feature importance for different feature combinations, with only the top ten features included in the thesis. These tables detail the importance values for each feature, helping us understand the predictive power of each model under various feature combinations.

4.2.5 32 Features

In this section, we delve into the performance of models using 32 features for predicting glioblastoma (GBM) recurrence patterns. The analysis is based on five-fold cross-validation, ensuring a comprehensive evaluation of the models' capabilities.

Table 1.16 32 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	95.45	95.24	76.19	90.48	100	91.47 $\pm$ 8.21
MRI Brain Tumor (Model B)	100	95.24	80.95	95.24	100	94.29 $\pm$ 7
MRI Ventricle (Model A)	100	100	95.24	95.24	90.48	96.19 $\pm$ 3.56
MRI Ventricle (Model B)	95.45	100	95.24	90.48	95.24	95.28 $\pm$ 3.01
MRI Tumor+Ventricle (Model A)	95.45	95.24	90.48	90.48	100	94.33 $\pm$ 3.57
MRI Tumor+Ventricle (Model B)	95.45	100	90.48	95.24	95.24	95.28 $\pm$ 3.01

Table 1.16 presents the validation accuracies for each fold, along with the average accuracy and standard deviation for each feature set.

According to the data in Tables 1.17 and 1.18, Model A achieved an average accuracy of $91.47\% \pm 8.21\%$ using brain tumor features, while Model B achieved an average accuracy of $94.29\% \pm 7\%$. This indicates that Model B performs better than Model A in predicting outcomes using brain tumor features.

Table 1.17 MRI Brain Tumor (Model A) 32 Features (Top 10)

Feature	Importance
flair-tumor-wavelet-LHL_glszm_SizeZoneNonUniformity	0.923680559
t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.795052146
t1-tumor-original_glrml_LongRunLowGrayLevelEmphasis	0.674255829
t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized	0.602573414
t1ce-tumor-original_firstorder_10Percentile	0.568000102
t1-tumor-wavelet-HLH_ngtdm_Busyness	0.562669564
flair-tumor-wavelet-HLL_firstorder_Kurtosis	0.535370753
t1-tumor-wavelet-LLH_ngtdm_Complexity	0.522829599
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	0.510415713
t1-tumor-wavelet-LHH_gldm_DependenceEntropy	0.502911976

Table 1.18 MRI Brain Tumor (Model B) 32 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.931845
t1-tumor-original_glrml_LongRunLowGrayLevelEmphasis	0.803496
flair-tumor-wavelet-LHL_glszm_SizeZoneNonUniformity	0.772203
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	0.738026
t2-tumor-wavelet-LLL_glcm_Imc2	0.713027
t1-tumor-original_glszm_SmallAreaHighGrayLevelEmphasis	0.622297
t1-tumor-wavelet-LHH_glszm_SmallAreaEmphasis	0.601376
t1ce-tumor-original_firstorder_10Percentile	0.589803
flair-tumor-lbp-3D-k_glrml_ShortRunLowGrayLevelEmphasis	0.579958
t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized	0.573642

Feature importance analysis for Model A shows that the most important brain tumor features include *flair-tumor-wavelet-LHL_glszm_SizeZoneNonUniformity*, *t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized*, and *t1-tumor-original_glrIm_LongRunLowGrayLevelEmphasis*. These features reflect the heterogeneity of the tumor texture and the non-uniformity in the distribution of gray-level values, which help distinguish morphological characteristics of the tumor, thereby improving prediction accuracy. For Model B, similar features also exhibit high importance, emphasizing the critical role of texture features in tumor classification.

In the MRI ventricle feature set, Model A achieved an average accuracy of $96.19\% \pm 3.56\%$, while Model B achieved an average accuracy of $95.28\% \pm 3.01\%$.

This indicates that ventricle features are also crucial in predicting tumor recurrence patterns. Important features include *t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis* and *flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis*. These features describe the textural details and gray-level distribution within the ventricles, reflecting the impact of the tumor on ventricular structures.

When combining MRI brain tumor and ventricle features, Model A achieved an average accuracy of $94.33\% \pm 3.57\%$, while Model B achieved

an average accuracy of $95.28\% \pm 3.01\%$. Feature importance analysis indicates that considering both brain tumor and ventricle features provides a more comprehensive reflection of the tumor's impact.

Table 1.19 MRI Ventricle (Model A) 32 Features (Top 10)

Feature	Importance
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	0.950746604
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	0.889573747
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	0.819843524
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	0.655488283
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.647201348
t1-ventricle-wavelet-HHL_glcm_InverseVariance	0.646614864
t1-ventricle-wavelet-HLL_gldm_DependenceVariance	0.624633606
flair-ventricle-wavelet-HHL_firstorder_Median	0.614478226
flair-ventricle-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	0.612463604
t2-ventricle-wavelet-LLH_glcm_InverseVariance	0.608003745

Table 1.20 MRI Ventricle (Model B) 32 Features (Top 10)

Feature	Importance
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	1.109123924
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	1.019843134
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	0.993672705
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	0.804770264
flair-ventricle-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	0.741330659
t1-ventricle-wavelet-HHL_glcm_InverseVariance	0.735551364
flair-ventricle-lbp-3D-m2_firstorder_Range	0.723709204
flair-ventricle-wavelet-HHL_firstorder_Median	0.701985523
t2-ventricle-wavelet-LLH_glcm_InverseVariance	0.653672954
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.653257157

Features such as *t1-tumor-wavelet-HLH_ngtdm_Busyness* and *flair-ventricle-wavelet-HHL_glszm_LowGrayLevelZoneEmphasis* show significant discriminative power across different MRI sequences, helping to improve prediction accuracy.

Table 1.21 MRI Tumor + Ventricle (Model A) 32 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-HLH_ngtdm_Busyness	0.80296525
t1-tumor-wavelet-LLH_ngtdm_Complexity	0.676382552
flair-tumor-wavelet-LHL_glszm_SizeZoneNonUniformity	0.667768747
flair-ventricle-wavelet-HHL_glszm_LowGrayLevelZoneEmphasis	0.605842311
t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.585136353
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	0.576470666
t1-tumor-original_glrml_LongRunLowGrayLevelEmphasis	0.53385521
flair-ventricle-lbp-3D-m1_firstorder_Skewness	0.522148499
t1-ventricle-wavelet-LHH_glcm_Idm	0.521742533
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.508167164

Table 1.22 MRI Tumor + Ventricle (Model B) 32 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-HLH_ngtdm_Busyness	0.943319744
flair-ventricle-wavelet-HHL_glszm_LowGrayLevelZoneEmphasis	0.742242164
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	0.720063254
t1-tumor-wavelet-LLH_ngtdm_Complexity	0.71145215
t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.707627374
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.676119653
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	0.669625456
flair-tumor-wavelet-LHH_glszm_SizeZoneNonUniformityNormalized	0.661567142
flair-ventricle-lbp-3D-m1_firstorder_Skewness	0.639125536
t1-ventricle-wavelet-LHH_glcm_Idm	0.623194

Tables 1.17 to 1.22 display the feature importance for different feature combinations, presenting the top ten features for each combination. These tables provide a detailed overview of the importance values for each feature, helping us understand the significance of each feature in the model predictions. It is demonstrated that combining brain tumor and ventricle features generally improves the accuracy of both models. These models' stability and predictive power are enhanced by the multi-faceted feature combinations, as evidenced by the detailed feature importance values. This

analysis underscores the value of integrating various feature types to enhance the robustness and accuracy of glioblastoma recurrence predictions.

4.2.6 64 Features

In this section, we analyze the results of model predictions using 64 features. According to the data in Table 1.23, the average accuracies for Model A and Model B with different feature combinations are as follows: for MRI Brain Tumor features, Model A achieved an average accuracy of 96.19%, while Model B achieved 95.28%. This indicates that Model A performs slightly better when predicting with brain tumor features. For MRI Ventricle features, both Model A and Model B achieved an average accuracy of 98.10%, demonstrating their high predictive power for ventricle features. When using the combined MRI Tumor+Ventricle features, Model A achieved an average accuracy of 95.28%, while Model B achieved 96.23%.

Table 1.23 64 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	100	95.24	90.48	95.24	100	96.19 $\pm$ 3.56
MRI Brain Tumor (Model B)	95.45	95.24	90.48	95.24	100	95.28 $\pm$ 3.01

MRI Ventricle (Model A)	100	100	90.48	100	100	98.10 $\pm$ 3.81
MRI Ventricle (Model B)	100	95.24	95.24	100	100	98.10 $\pm$ 2.33
MRI Tumor+Ventricle (Model A)	95.45	95.24	90.48	95.24	100	95.28 $\pm$ 3.01
MRI Tumor+Ventricle (Model B)	95.45	95.24	95.24	95.24	100	96.23 $\pm$ 1.88

For the brain tumor features in Model A, the most important features include *t1-tumor-wavelet-LHH_firstorder_Kurtosis*, *t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized*, and *t2-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized*. These features reflect the complexity and variability within the T1 and T2 weighted images, which are crucial for predicting tumor characteristics and recurrence patterns. In Model B, the top features are similar but show slight differences in their importance values. The features *t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized* and *t2-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized* again appear at the top, indicating their strong relevance across different models.

Table 1.24 MRI Brain Tumor (Model A) 64 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-LHH_firstorder_Kurtosis	0.481700314
t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized	0.47473335
t2-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.465758548
t1ce-tumor-wavelet-LLL_firstorder_RobustMeanAbsoluteDeviation	0.446434806
t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.404468802

flair-tumor-wavelet-LHH_glszm_GrayLevelNonUniformity	0.395008855
t1-tumor-original_glrml_LongRunLowGrayLevelEmphasis	0.394889598
flair-tumor-lbp-3D-k_glrml_LowGrayLevelRunEmphasis	0.372120035
t1-tumor-wavelet-LLH_ngtdm_Complexity	0.35277658
t1-tumor-wavelet-HLL_glszm_LowGrayLevelZoneEmphasis	0.352756635

Table 1.25 MRI Brain Tumor (Model B) 64 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized	0.586882407
t2-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.557819523
t1-tumor-wavelet-LHH_firstorder_Kurtosis	0.542548489
flair-tumor-wavelet-LHH_glszm_GrayLevelNonUniformity	0.472578877
t1ce-tumor-wavelet-LLL_firstorder_RobustMeanAbsoluteDeviation	0.467342019
t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized	0.466485634
flair-tumor-lbp-3D-k_glrml_LowGrayLevelRunEmphasis	0.418985065
t1-tumor-wavelet-LHH_glszm_SmallAreaEmphasis	0.413559506
t1-tumor-original_glrml_LongRunLowGrayLevelEmphasis	0.390406238
t1ce-tumor-original_shape_Elongation	0.385671364

For ventricle features, both models highlight the importance of *flair-ventricle-lbp-3D-m2_firstorder_Range* and *flair-ventricle-wavelet-HHL_glszm_ZonePercentage*, which measure the texture and structural details of the ventricles, aiding in the prediction of morphological changes associated with tumor development.

Table 1.26 MRI Ventricle (Model A) 64 Features (Top 10)

Feature	Importance
flair-ventricle-lbp-3D-m2_firstorder_Range	0.47438433
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	0.471851708
flair-ventricle-wavelet-HLH_glszm_SizeZoneNonUniformityNormalized	0.451455242
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.439256678
t1-ventricle-wavelet-LHL_ngtdm_Contrast	0.411314192
flair-ventricle-wavelet-HHL_glszm_LowGrayLevelZoneEmphasis	0.411073577
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	0.401355162
t1ce-ventricle-original_gldm_DependenceNonUniformity	0.395601948
flair-ventricle-lbp-3D-k_glrml_ShortRunLowGrayLevelEmphasis	0.394672864
t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis	0.392333031

Table 1.27 MRI Ventricle (Model B) 64 Features (Top 10)

Feature	Importance
flair-ventricle-lbp-3D-m2_firstorder_Range	0.563556684
flair-ventricle-wavelet-HHL_glszm_ZonePercentage	0.534461773
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.509124235
flair-ventricle-lbp-3D-k_glrlm_ShortRunLowGrayLevelEmphasis	0.495035299
flair-ventricle-wavelet-HLH_glszm_SizeZoneNonUniformityNormalized	0.475195214
t1-ventricle-lbp-3D-k_firstorder_Minimum	0.466560592
flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis	0.453178352
t1-ventricle-lbp-3D-m1_glszm_ZoneEntropy	0.453140678
flair-ventricle-wavelet-LLL_glcmm_ClusterProminence	0.449115327
t1-ventricle-lbp-3D-m1_glszm_SizeZoneNonUniformityNormalized	0.446157745

When combining brain tumor and ventricle features, the models show a mix of high-importance features from both datasets. For Model A, key features include *t1-tumor-wavelet-HLL_glszm_LowGrayLevelZoneEmphasis*, *flair-tumor-wavelet-LHL_glszm_GrayLevelNonUniformity*, and *t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis*. These features capture the intricate spatial relationships and textures within both brain tumor and ventricle images. In Model B, combined features such as *t2-ventricle-wavelet-HHH_glrlm_ShortRunLowGrayLevelEmphasis* and *flair-ventricle-lbp-3D-m2_firstorder_Range* are prominent, showing the model's reliance on detailed texture and structural measurements from both the tumor and ventricle images.

Table 1.28 MRI Tumor + Ventricle (Model A) 64 Features (Top 10)

Feature	Importance
t1-tumor-wavelet-HLL_glszm_LowGrayLevelZoneEmphasis	0.413344286
flair-tumor-wavelet-LHL_glszm_GrayLevelNonUniformity	0.410928844
t2-ventricle-wavelet-HHH_glrlm_ShortRunLowGrayLevelEmphasis	0.39863735
t1-tumor-wavelet-HLH_ngtdm_Busyness	0.394733119
flair-ventricle-wavelet-HLH_glszm_SizeZoneNonUniformityNormalized	0.381532716

t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized	0.365671513
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.362136781
flair-ventricle-lbp-3D-m2_firstorder_Range	0.360591775
t1-tumor-wavelet-LLL_glcm_ClusterProminence	0.356513593
t1ce-tumor-wavelet-LLL_firstorder_RobustMeanAbsoluteDeviation	0.343263803

Table 1.29 MRI Tumor + Ventricle (Model B) 64 Features (Top 10)

Feature	Importance
t2-ventricle-wavelet-HHH_glrml_ShortRunLowGrayLevelEmphasis	0.487862506
flair-ventricle-lbp-3D-m2_firstorder_Range	0.46303418
flair-tumor-wavelet-LHL_glszm_GrayLevelNonUniformity	0.450730795
t1-tumor-wavelet-HLL_glszm_LowGrayLevelZoneEmphasis	0.449586148
flair-ventricle-lbp-3D-m2_firstorder_Maximum	0.4120072
t1-ventricle-lbp-3D-m1_glszm_ZoneEntropy	0.406447785
t1-tumor-wavelet-HHL_glszm_SizeZoneNonUniformityNormalized	0.403714469
flair-ventricle-wavelet-HLH_glszm_SizeZoneNonUniformityNormalized	0.403345315
flair-tumor-wavelet-LHH_glszm_GrayLevelNonUniformity	0.403127371
t1ce-tumor-wavelet-LLL_firstorder_RobustMeanAbsoluteDeviation	0.398615306

Tables illustrating the feature importance for each model and feature combination (Table 1.23 to 1.28) showcase the top ten important features included in the thesis. These tables help visualize the relative importance of different features in the model predictions, further enhancing our understanding of the relationship between feature selection and model performance.

4.2.7 128 Features

This section details the performance of using 128 feature combinations in classification accuracy. Each feature combination includes 128 features and undergoes five-fold cross-validation.

Table 1.30 128 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	95.45	95.24	100	90.48	100	96.23 $\pm$ 3.55
MRI Brain Tumor (Model B)	90.91	90.48	90.48	95.24	100	93.42 $\pm$ 3.75
MRI Ventricle (Model A)	100	100	85.71	100	95.24	96.19 $\pm$ 5.55
MRI Ventricle (Model B)	90.91	95.24	100	90.48	100	95.33 $\pm$ 4.16
MRI Tumor+Ventricle (Model A)	100	90.48	95.24	100	100	97.14 $\pm$ 3.81
MRI Tumor+Ventricle (Model B)	100	95.24	95.24	100	100	98.1 $\pm$ 2.33

Table 1.30 shows the specific accuracy and average accuracy of these feature combinations in five-fold cross-validation. The results indicate that the combined MRI brain tumor and ventricle features perform the best, followed by using brain tumor features alone and ventricle features alone. This suggests that with 128 features, combining features can significantly enhance the model's classification accuracy.

4.2.8 256 Features

This section details the performance of using 256 feature combinations

in classification accuracy. Each feature combination includes 256 features and undergoes five-fold cross-validation.

Table 1.31 256 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	95.45	95.24	100	90.48	100	96.23 ± 3.55
MRI Brain Tumor (Model B)	86.36	95.24	100	90.48	100	94.42 ± 5.36
MRI Ventricle (Model A)	95.45	100	95.24	100	100	98.14 ± 2.28
MRI Ventricle (Model B)	90.91	100	85.71	100	100	95.32 ± 5.96
MRI Tumor+Ventricle (Model A)	95.45	95.24	100	100	100	98.14 ± 2.28
MRI Tumor+Ventricle (Model B)	100	95.24	100	100	100	99.05 ± 1.9

Table 1.31 shows the specific accuracy and average accuracy of these feature combinations in five-fold cross-validation. The results indicate that the combined MRI brain tumor and ventricle features perform the best, with the accuracy of using brain tumor features alone and ventricle features alone also being quite high. This suggests that with 256 features, combining features can significantly enhance the model's classification accuracy, and the stability of the model is also improved.

4.2.9 512 Features

Table 1.32 512 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	86.36	95.24	85.71	85.71	100	90.60 $\pm$ 5.93
MRI Brain Tumor (Model B)	90.91	95.24	90.48	85.71	100	92.47 $\pm$ 4.83
MRI Ventricle (Model A)	86.36	95.24	85.71	95.24	95.24	91.56 $\pm$ 4.51
MRI Ventricle (Model B)	77.27	95.24	85.71	95.24	95.24	89.74 $\pm$ 7.25
MRI Tumor+Ventricle (Model A)	95.45	95.24	100	100	100	98.14 $\pm$ 2.28
MRI Tumor+Ventricle (Model B)	95.45	95.24	100	100	100	98.14 $\pm$ 2.28

This section details the performance of using 512 feature combinations in classification accuracy. Each feature combination includes 512 features and undergoes five-fold cross-validation. Table 1.32 shows the specific accuracy and average accuracy of these feature combinations in five-fold cross-validation. The results indicate that the combined MRI brain tumor and ventricle features have the highest accuracy while using brain tumor features and ventricle features alone yields lower accuracy. This suggests that with 512 features, combining features can significantly enhance the model's classification accuracy, and the model's stability is also improved.

However, increasing the number of features may introduce the risk of overfitting, which requires careful adjustment.

4.2.10 1024 Features

In this section, we evaluate the performance of models utilizing 1024 features. Each feature combination undergoes five-fold cross-validation to ensure the robustness of the results. Table 1.33 presents the detailed accuracy results for each fold, along with the average accuracy and standard deviation for the various feature sets.

The results indicate that the combined MRI brain tumor and ventricle features outperform other feature sets, with Model A achieving an average accuracy of $97.23\% \pm 3.66\%$ and Model B achieving $96.28\% \pm 3.43\%$. This suggests that integrating a larger number of features from both the brain tumor and ventricle significantly enhances the model's predictive performance.

Table 1.33 1024 Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	77.27	95.24	71.43	90.48	95.24	85.93 ± 9.79
MRI Brain Tumor (Model B)	68.18	85.71	66.67	85.71	90.48	79.35 ± 9.9

MRI Ventricle (Model A)	68.18	80.95	76.19	90.48	90.48	81.26 $\pm$ 8.57
MRI Ventricle (Model B)	63.64	85.71	80.95	90.48	90.48	82.25 $\pm$ 9.95
MRI Tumor+Ventricle (Model A)	90.91	95.24	100	100	100	97.23 $\pm$ 3.66
MRI Tumor+Ventricle (Model B)	90.91	95.24	95.24	100	100	96.28 $\pm$ 3.43

For the brain tumor feature set alone, Model A achieves an average accuracy of 85.93% $\pm$ 9.79%, while Model B achieves 79.35% $\pm$ 9.9%. Similarly, for the ventricle feature set alone, Model A achieves 81.26% $\pm$ 8.57%, and Model B achieves 82.25% $\pm$ 9.95%. These results highlight that while using a large number of features improves performance, combining features from different regions (brain tumor and ventricle) leads to even better outcomes.

However, the high accuracy observed in some folds, particularly for the combined feature sets, raises the concern of potential overfitting. Overfitting occurs when the model captures noise or random fluctuations in the training data, leading to high performance on the training data but poor generalization to new, unseen data. The relatively high standard deviations observed, particularly in the brain tumor and ventricle feature sets, further suggest variability in model performance across different folds, which could be indicative of overfitting.

4.2.11 All Features

In this section, we analyze the performance of classification models using all available features. Each model underwent five-fold cross-validation to ensure comprehensive evaluation. Table 1.34 summarizes the validation accuracy for each fold and provides the average accuracy and standard deviation for each feature set.

Table 1.34 All Features

Feature Set	Valid Acc Fold 1(%)	Valid Acc Fold 2(%)	Valid Acc Fold 3(%)	Valid Acc Fold 4(%)	Valid Acc Fold 5(%)	Average $\pm$ std(%)
MRI Brain Tumor (Model A)	54.55	57.14	52.38	80.95	61.90	61.38 $\pm$ 10.28
MRI Brain Tumor (Model B)	54.55	52.38	52.38	57.14	61.90	55.67 $\pm$ 3.57
MRI Ventricle (Model A)	54.55	66.67	47.62	76.19	76.19	64.24 $\pm$ 11.5
MRI Ventricle (Model B)	50	61.90	47.62	85.71	76.19	64.28 $\pm$ 14.75
MRI Tumor+Ventricle (Model A)	54.55	61.90	33.33	66.67	71.43	57.58 $\pm$ 13.34
MRI Tumor+Ventricle (Model B)	59.09	66.67	33.33	76.19	61.90	59.44 $\pm$ 14.29

The results show varying performance across different feature sets when using all features. For the MRI brain tumor feature set, Model A achieved an average accuracy of 61.38% $\pm$ 10.28%, while Model B

achieved $55.67\% \pm 3.57\%$. These results indicate that including all brain tumor data features does not necessarily improve performance and may adversely affect model accuracy.

For the MRI ventricle feature set, both Model A and Model B performed similarly, with average accuracies of $64.24\% \pm 11.5\%$ and $64.28\% \pm 14.75\%$, respectively. However, the relatively high standard deviation suggests variability in model performance across folds, potentially indicating overfitting or the presence of irrelevant features.

When combining MRI brain tumor and ventricle features, the performance of both models did not significantly improve. Model A had an average accuracy of $57.58\% \pm 13.34\%$, and Model B had $59.44\% \pm 14.29\%$. These results indicate that simply combining all features from both datasets is not the optimal approach and may reduce model performance due to overfitting and the inclusion of irrelevant or redundant features.

The results suggest that increasing the number of features does not necessarily enhance the model's learning ability. Instead, it may cause the model to learn irrelevant features, leading to decreased performance. The high variability and standard deviation in validation accuracy further emphasize the risk of overfitting when using all features.

Chapter 5 Discussion

The experimental results from our neural network model for predicting GBM recurrence patterns using radiomic features from multimodal MRI images highlight critical insights and clinical implications. Our findings demonstrate the importance of feature selection and combining multiple anatomical and functional aspects of the brain in predictive modeling.

Combining brain tumor and ventricle features consistently enhanced the prediction accuracy of GBM recurrence patterns. This result underscores the necessity of incorporating comprehensive information that captures the intricate interactions between different brain structures. Models utilizing combined features outperformed those relying solely on brain tumor or ventricle features, suggesting the interplay between the tumor and the surrounding brain environment is pivotal for accurate predictions.

Feature importance analysis revealed key attributes in predicting GBM recurrence. For brain tumor features, important attributes included *flair-tumor-wavelet-LHL_glszm_SizeZoneNonUniformity* and *t1-tumor-wavelet-LHH_gldm_DependenceNonUniformityNormalized*, which reflects tumor texture heterogeneity and gray-level value distribution. For ventricle features, *t2-ventricle-wavelet-HHH_gldm_LowGrayLevelEmphasis* and *flair-ventricle-wavelet-HLH_glszm_SmallAreaEmphasis* were significant,

highlighting the impact of ventricle texture details on model performance.

Our findings indicate that increasing the number of features does not necessarily improve model performance. Using many features (e.g., 1024 or all available features) led to decreased accuracy and increased variability due to overfitting. This emphasizes the risks associated with overfitting and including irrelevant or redundant features.

Clinically, accurately predicting GBM recurrence patterns has significant implications. Early and precise predictions can guide personalized treatment plans, improve patient outcomes, and optimize medical resource allocation. Integrating radiomic features from brain tumors and ventricles offers a comprehensive view of tumor behavior and its interaction with surrounding brain structures, enhancing the precision of oncological assessments and treatment planning.

Chapter 6 Conclusion

This study demonstrates the efficacy of combining radiomic features from brain tumors and ventricles in predicting GBM recurrence patterns, underscoring the critical importance of feature selection. Utilizing a moderate number of carefully selected features significantly enhances model performance. The high accuracy achieved by this combined approach highlights its potential for clinical application, paving the way for more personalized and effective treatment strategies for glioblastoma patients.

Furthermore, integrating ventricular features with brain tumor characteristics provides a comprehensive understanding of the tumor's impact on brain physiology. This holistic approach not only improves the accuracy of predictive models but also facilitates the development of tailored treatment plans, significantly enhancing patient care and outcomes.

However, this study has certain limitations. The dataset is limited to preoperative MRI scans from a single institution (Keelung Chang Gung Memorial Hospital). Future research should include multi-institutional datasets to validate the model's generalizability across diverse populations and imaging protocols. Integrating clinical data, such as patient demographics, genetic markers, and treatment histories, with radiomic features could yield a more accurate prediction model. Future studies

should aim to develop such integrative models to fully leverage the potential of artificial intelligence in clinical decision-making.

Additionally, incorporating temporal data to predict recurrence patterns and the timing of recurrence can provide valuable insights for clinical decision-making. Understanding the temporal dynamics of tumor recurrence can help clinicians plan more effective follow-up schedules and intervention strategies.

In the future, an end-to-end approach can be employed to directly predict GBM recurrence patterns from MRI data without manual intervention, such as image segmentation. This approach has significant potential for practical clinical use, allowing raw MRI images to be directly input into the model to obtain predictive results without requiring manual image segmentation. Future work should focus on implementing this end-to-end pipeline and integrating clinical data to predict other outcomes, further enhancing the practicality and applicability of our methods in clinical settings.

In summary, our research demonstrates the potential of combining radiomic features from brain tumors and ventricles to enhance the prediction of GBM recurrence patterns. The findings emphasize the importance of feature selection and integrating comprehensive anatomical and functional information. Future research should focus on validating

these results across diverse datasets, exploring additional brain regions, incorporating temporal dynamics, and using end-to-end methods to further improve predictive accuracy and clinical utility.



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